

Cancer Statistics for African Americans, 2016: Progress and Opportunities in Reducing Racial Disparities

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In this article, the American Cancer Society provides the estimated number of new cancer cases and deaths for blacks in the United States and the most recent data on cancer incidence, mortality, survival, screening, and risk factors for cancer. Incidence data are from the National Cancer Institute, the Centers for Disease Control and Prevention, and the North American Association of Central Cancer Registries, and mortality data are from the National Center for Health Statistics. Approximately 189,910 new cases of cancer and 69,410 cancer deaths will occur among blacks in 2016. Although blacks continue to have higher cancer death rates than whites, the disparity has narrowed for all cancers combined in men and women and for lung and prostate cancers in men. In contrast, the racial gap in death rates has widened for breast cancer in women and remained level for colorectal cancer in men. The reduction in overall cancer death rates since the early 1990s translates to the avoidance of more than 300,000 deaths among blacks. In men, incidence rates from 2003 to 2012 decreased for all cancers combined (by 2.0% per year) as well as for the top 3 cancer sites (prostate, lung, and colorectal). In women, overall rates during the corresponding time period remained unchanged, reflecting increasing trends in breast cancer combined with decreasing trends in lung and colorectal cancer rates. Five-year relative survival is lower for blacks than whites for most cancers at each stage of diagnosis. The extent to which these disparities reflect unequal access to health care versus other factors remains an active area of research. Progress in reducing cancer death rates could be accelerated by ensuring equitable access to prevention, early detection, and high-quality treatment. *CA Cancer J Clin* 2016;66:290-308. © 2016 American Cancer Society.

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Introduction

In the United States, African Americans bear a disproportionate share of the cancer burden, having the highest death rate and shortest survival of any racial or ethnic group for most cancers. The causes of these inequalities are complex and reflect social and economic disparities more than biological differences. For example, in 2014, 26% of blacks, compared with 10% of non-Hispanic whites, were living below the federal poverty level, and 22% of blacks had completed 4 years of college compared with 36% of non-Hispanic whites.^{1,2} Persons with lower socioeconomic status are more likely to engage in behaviors that increase cancer risk, in part because of marketing strategies that target these populations as well as environmental and community factors, such as fewer opportunities for physical activity and less access to fresh fruits and vegetables. We recognize that race is a social construct; however, because much US health data are reported by race, racial classification remains useful for describing general patterns of health within the nation. Although we use the terms African Americans and blacks interchangeably, the data provided herein are for those identified by black race and exclude those of Hispanic ethnicity when possible. A report on cancer statistics for Hispanics was published previously in this journal.³ Herein, we provide cancer incidence, survival, and mortality statistics for blacks, including the estimated numbers of new cases and deaths in 2016, as well as the prevalence of cancer risk factors and screening uptake. We also estimate the total number of cancer deaths averted among blacks as a result of the decline in cancer death rates since the early 1990s.

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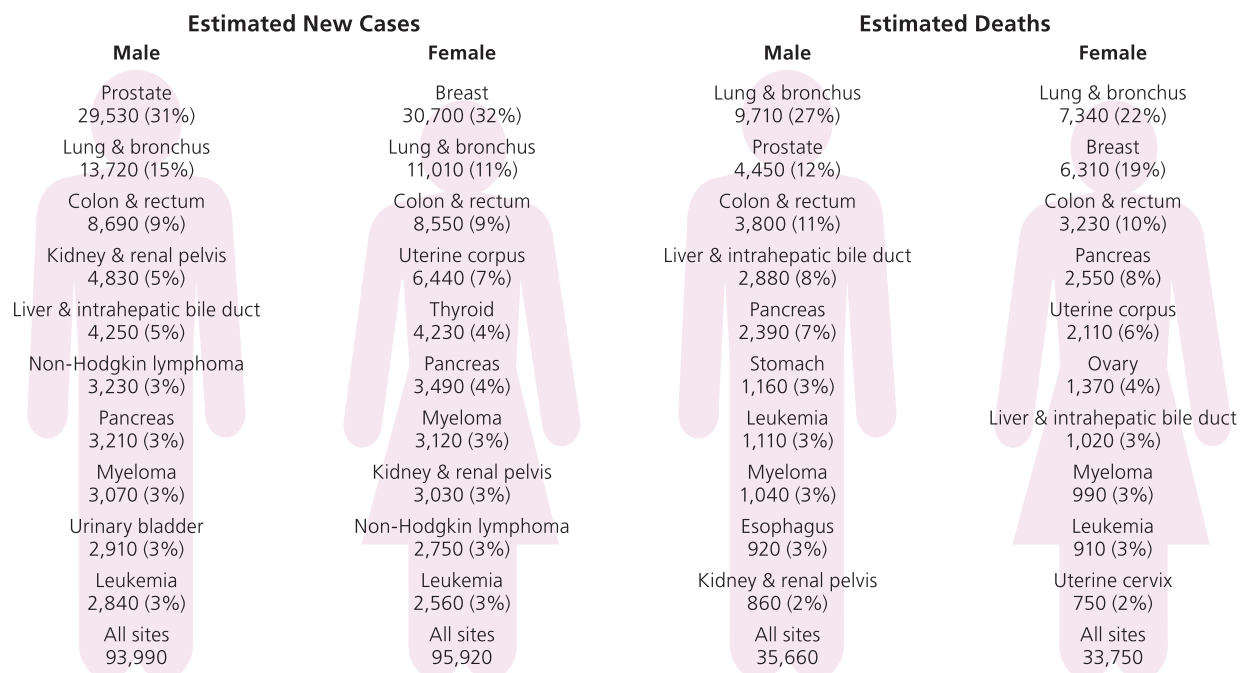


FIGURE 1. Leading Sites of New Cancer Cases and Deaths Among Blacks, 2016 Estimates.

Estimates are rounded to the nearest 10 cell and exclude basal cell and squamous cell skin cancers and in situ carcinoma except urinary bladder.

Materials and Methods

Incidence and Mortality Data

There are two sources for cancer incidence data reported in this article. The Surveillance, Epidemiology, and End Results (SEER) program of the National Cancer Institute (NCI) reports long-term, high-quality, population-based incidence data covering up to 28% of the US population. Long-term incidence trends (1975–2012) were based upon data from the SEER 9 registries (Connecticut, Hawaii, Iowa, New Mexico, Utah, and the metropolitan areas of Atlanta, Detroit, San Francisco–Oakland, and Seattle–Puget Sound), representing approximately 9% of the US population.⁴ As of 1992, SEER data have been available for 4 additional SEER registries (Alaska Natives, Los Angeles County, San Jose–Monterey, and rural Georgia), which increase coverage of minority groups, allowing for stratification by race and ethnicity. Delay-adjusted data from these (SEER 13) registries, which represent 14% of the US population, were the source for the annual percent change in incidence from 2003 to 2012.⁵ Five-year relative survival rates, stage at diagnosis, and the lifetime probability of developing cancer were based upon data from the SEER 18 registries, which also include data from Greater California, Greater Georgia, Kentucky, Louisiana, and New Jersey and are available from 2000 onward.⁶ The probability of developing cancer was calculated using NCI's DevCan software

(version 6.7.3).⁷ Much of the statistical information presented here was previously published in the *SEER Cancer Statistics Review 1975–2012*.⁸

The North American Association of Central Cancer Registries (NAACCR) compiles and reports incidence data for 1995 forward from cancer registries that participate in the SEER program or from the Centers for Disease Control and Prevention's National Program of Cancer Registries. These data approach 100% coverage of the US population in the most recent time period and were the source for the projected new cancer cases in 2016 and incidence rates in the most recent time period (2008–2012).⁹ Overall US rates based on data from NAACCR include all states except Arkansas, Minnesota, and Nevada, because these states did not submit data or failed to meet NAACCR high-quality standards for 1 or more years during 2008 through 2012. Some of the data presented here were previously published in volumes 1 and 2 of *Cancer in North America: 2008–2012*.^{10,11}

Mortality data were obtained from the National Center for Health Statistics as reported by the SEER program.¹² Data are available for whites and blacks beginning in 1969 and by Hispanic ethnicity since 1990. When available, data are presented for non-Hispanic blacks and non-Hispanic whites. All cancer cases and deaths were accessed using SEER*Stat software (version 8.2.1).¹³ Population data were obtained from the US Census Bureau. Incidence and death rates were age-standardized to the 2000 US standard population and expressed per 100,000 persons.

TABLE 1. Lifetime Probability (%) of Developing or Dying From Invasive Cancers by Race/Ethnicity and Sex, United States, 2010-2012*

		DEVELOPING		DYING	
		BLACK	NH WHITE	BLACK	NH WHITE
All sites†	Male	40.8 (1 in 2)	42.4 (1 in 2)	23.4 (1 in 4)	22.8 (1 in 4)
	Female	34.3 (1 in 3)	39.0 (1 in 3)	19.4 (1 in 5)	19.5 (1 in 5)
Prostate	Male	18.2 (1 in 6)	13.3 (1 in 8)	4.4 (1 in 23)	2.4 (1 in 42)
Breast	Female	11.1 (1 in 9)	13.1 (1 in 8)	3.3 (1 in 31)	2.7 (1 in 37)
Lung & bronchus	Male	7.5 (1 in 13)	7.5 (1 in 13)	6.4 (1 in 16)	6.6 (1 in 15)
	Female	5.4 (1 in 19)	6.7 (1 in 15)	4.2 (1 in 24)	5.3 (1 in 19)
Colon & rectum	Male	4.9 (1 in 21)	4.6 (1 in 22)	2.4 (1 in 42)	1.9 (1 in 52)
	Female	4.7 (1 in 21)	4.3 (1 in 23)	2.1 (1 in 47)	1.8 (1 in 56)
Uterine corpus	Female	2.5 (1 in 39)	2.9 (1 in 35)	0.9 (1 in 108)	0.5 (1 in 184)
Kidney	Male	2.0 (1 in 51)	2.1 (1 in 48)	0.5 (1 in 204)	0.6 (1 in 158)
	Female	1.3 (1 in 79)	1.2 (1 in 83)	0.3 (1 in 328)	0.3 (1 in 288)
Urinary bladder	Male	1.9 (1 in 54)	4.4 (1 in 23)	0.5 (1 in 194)	1.0 (1 in 100)
	Female	0.8 (1 in 124)	1.3 (1 in 79)	0.4 (1 in 285)	0.4 (1 in 284)
Pancreas	Male	1.5 (1 in 67)	1.5 (1 in 65)	1.4 (1 in 74)	1.4 (1 in 72)
	Female	1.7 (1 in 58)	1.4 (1 in 69)	1.5 (1 in 66)	1.3 (1 in 76)
Non-Hodgkin lymphoma	Male	1.4 (1 in 70)	2.5 (1 in 40)	0.5 (1 in 201)	0.9 (1 in 110)
	Female	1.2 (1 in 84)	2.0 (1 in 50)	0.4 (1 in 239)	0.7 (1 in 140)
Uterine cervix	Female	0.8 (1 in 130)	0.6 (1 in 176)	0.4 (1 in 265)	0.2 (1 in 506)
Thyroid	Male	0.3 (1 in 368)	0.7 (1 in 148)	<0.1 (1 in 2,908)	0.1 (1 in 1,783)
	Female	1.0 (1 in 98)	1.9 (1 in 54)	0.1 (1 in 1,556)	0.1 (1 in 1,581)
Liver & bile duct	Male	1.5 (1 in 69)	1.0 (1 in 99)	1.1 (1 in 88)	0.8 (1 in 123)
	Female	0.5 (1 in 195)	0.4 (1 in 249)	0.5 (1 in 193)	0.4 (1 in 237)
Leukemia	Male	1.14 (1 in 88)	1.84 (1 in 53)	0.7 (1 in 147)	1.1 (1 in 92)
	Female	0.92 (1 in 109)	1.28 (1 in 77)	0.6 (1 in 176)	0.7 (1 in 134)

NH indicates non-Hispanic.

*For people who have not been previously diagnosed with cancer.

†All sites excludes basal cell and squamous cell skin cancers and in situ cancers except urinary bladder.

Note: Percentages and "1 in" numbers may not be equivalent due to rounding.

Source: DevCan: Probability of Developing or Dying of Cancer Software, Version 6.7.3.⁷

All cancer cases were classified according to the International Classification of Diseases for Oncology.¹⁴ Causes of death were classified according to the International Classification of Diseases.¹⁵ The annual percent change in rates was quantified using NCI's Joinpoint Regression Program (version 4.2.0.2).¹⁶

Projected Cancer Cases and Deaths in 2016

The most recent year for which incidence and mortality data are available lags 2 to 4 years behind the current year because of the time required for data collection, compilation, quality control, and dissemination. Therefore, we projected the numbers of new cancer cases and deaths for blacks in the United States in 2016 to provide an estimate of the contemporary cancer burden. The number of invasive cancer cases that occurred each year during 1998 through 2012 was estimated using a 3-step spatiotemporal model based on high-quality incidence data from 49 states and the District of Columbia, representing approximately 94% population coverage (data

were lacking for all years for Minnesota and for some years for other states). This method accounts for expected delays in case reporting and considers geographic variations in sociodemographic and lifestyle factors, medical settings, and cancer screening behaviors as predictors of incidence.¹⁷ Finally, a temporal projection method (vector autoregression) was applied to all 15 years of data to estimate counts for 2016. For complete details of the case projection methodology, please refer to Zhu et al.¹⁸

The number of cancer deaths expected to occur in 2016 among blacks in the United States was estimated based on the annual percent change in the actual number of cancer deaths from 1998 through 2012 as reported to the National Center for Health Statistics. For the complete details of this methodology, please refer to Chen et al.¹⁹

Estimated Cancer Deaths Averted

The estimated numbers of cancer deaths averted in black men and women because of the reduction in overall

TABLE 2. Comparison of Cancer Incidence Rates Between Non-Hispanic (NH) Blacks and Whites, United States, 2008–2012

MALE					FEMALE				
CANCER	NH BLACK RATE*	NH WHITE RATE*	ABSOLUTE DIFFERENCE†	RATE RATIO‡	CANCER	NH BLACK RATE*	NH WHITE RATE*	ABSOLUTE DIFFERENCE†	RATE RATIO‡
Kaposi sarcoma	1.7	0.5	1.2	3.57§	Kaposi sarcoma	0.2	<0.1	0.1	3.96§
Myeloma	14.8	7.0	7.8	2.11§	Myeloma	11.1	4.3	6.8	2.58§
Stomach	15.1	7.8	7.3	1.93§	Stomach	8.0	3.5	4.5	2.30§
Liver & IHB	16.5	9.3	7.2	1.77§	Liver & IHB	4.8	3.2	1.6	1.52§
Prostate	208.7	123.0	85.7	1.70§	Uterine cervix	10.0	7.1	2.9	1.41§
Larynx	9.3	6.3	3.0	1.48§	Pancreas	14.4	10.6	3.8	1.36§
Breast	2.0	1.4	0.6	1.45§	Esophagus	2.5	1.8	0.7	1.34§
Colon & rectum	60.3	47.4	12.9	1.27§	Colon & rectum	44.1	36.2	7.9	1.22§
Pancreas	17.2	14.0	3.2	1.23§	Kidney & renal pelvis	13.0	11.3	1.7	1.15§
Lung & bronchus	93.4	79.3	14.1	1.18§	Breast	124.3	128.1	−3.8	0.97§
Kidney & renal pelvis	24.2	21.8	2.4	1.11§	Uterine corpus	23.0	25.5	−2.5	0.90§
Hodgkin lymphoma	3.2	3.4	−0.2	0.95§	Hodgkin lymphoma	2.4	2.7	−0.3	0.88§
Esophagus	8.0	8.8	−0.8	0.90§	Lung & bronchus	51.4	58.7	−7.3	0.87§
Oral cavity & pharynx	15.3	18.1	−2.8	0.84§	Leukemia	8.6	10.7	−2.1	0.80§
Leukemia	13.2	17.7	−4.5	0.75§	Oral cavity & pharynx	5.2	6.7	−1.5	0.78§
Non-Hodgkin lymphoma	17.2	24.1	−6.9	0.71§	Ovary	9.6	12.4	−2.8	0.77§
Brain & ONS	4.9	8.8	−3.9	0.56§	Non-Hodgkin lymphoma	12.0	16.6	−4.6	0.72§
Urinary bladder	19.8	40.2	−20.4	0.49§	Urinary bladder	6.7	9.9	−3.2	0.68§
Thyroid	3.7	7.7	−4.0	0.48§	Thyroid	12.9	21.9	−9.0	0.59§
Testis	1.4	6.8	−5.4	0.21§	Brain & ONS	3.6	6.3	−2.7	0.58§
Melanoma of the skin	1.1	31.3	−30.2	0.04§	Melanoma of the skin	1.0	20.6	−19.6	0.05§
All sites	592.3	528.9	63.4	1.12§	All sites	408.1	436.2	−28.1	0.94§

IHB indicates intrahepatic bile duct; ONS, other nervous system.

*Rates are per 100,000 and age adjusted to the 2000 US standard population.

†The absolute difference is the rate in blacks minus the rate in whites.

‡The rate ratio is the unrounded rate in blacks divided by the unrounded rate in whites.

§The rate ratio is significantly different from one ($P < .05$).

Note: Sites are listed in descending order by rate ratio.

Source: North American Association of Central Cancer Registries.⁹

cancer death rates were determined by subtracting the number of recorded cancer deaths from the number that would have been expected if cancer death rates had remained at their peak. The expected numbers of deaths were calculated by applying the 5-year age-specific cancer death rates in the peak year for age-standardized cancer death rates (1990 in men, 1991 in women) to the corresponding age-specific populations in the subsequent years through 2012. We then summed the difference between the number of expected and observed deaths in each age group and calendar year for men and women separately.

Risk Factors and Screening Data

Data on behavioral risk factors (cigarette smoking, obesity, and physical inactivity) and receipt of cancer screening were obtained from two national surveys: the National Health Interview Survey (NHIS)²⁰ and the National Health and Nutrition Examination Survey (NHANES).²¹ NHANES is the preferred source of information for obesity prevalence in the United States, because height and

weight are measured rather than reported by participants. All surveys were analyzed using SUDAAN statistical software (version 11.0.1; RTI International, Research Triangle Park, NC) to obtain weighted prevalence estimates, which are considered representative of the noninstitutionalized civilian population.

Selected Findings

Overall Cancer Occurrence

Incidence

About 189,910 new cancer cases are expected to be diagnosed among blacks in 2016, including 93,990 cases in men and 95,920 cases in women (Fig. 1). Prostate cancer is expected to be the most commonly diagnosed cancer in men, and breast cancer is expected to be the most commonly diagnosed cancer in women. Cancers of the lung and colorectum will be the second-most and third-most commonly diagnosed cancers in both black men and black women. The four most common cancers (breast, prostate, colorectal, and lung) account for more than half of all

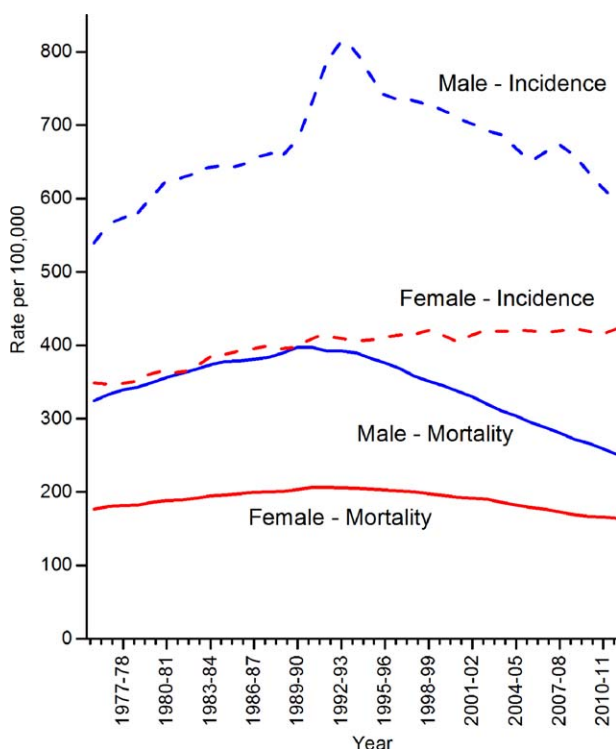


FIGURE 2. Trends in Cancer Incidence and Death Rates Among Blacks, United States, 1975 to 2012.

Rates are age adjusted, 2-year moving averages. Incidence rates are adjusted for reporting delay.

Sources: *Incidence*: Surveillance, Epidemiology, and End Results (SEER) Program, SEER 9 registries, National Cancer Institute.⁴ *Mortality*: National Center for Health Statistics, Centers for Disease Control and Prevention.¹²

cancer cases. The lifetime probability of being diagnosed with cancer among black men and black women is 41% and 34%, respectively, compared with 42% and 39%, respectively, among whites (Table 1).

Differences in cancer incidence rates between blacks and whites in the United States are described in Table 2. Among black males, incidence rates are higher for all cancers combined (12% higher) and for the most common cancers, including cancers of the prostate, lung, colorectum, kidney, and pancreas. In contrast, black females have a 6% lower overall incidence rate for all cancers combined and for many cancers, including lung, breast (only 3% lower), and uterine cancers.

Incidence rates for Kaposi sarcoma (KS), stomach cancer, and multiple myeloma are about 2 to 4 times higher in blacks than in whites (Table 2). Although it is a relatively rare cancer, incidence rates of KS are 4 times higher in black men and women than in their white counterparts. In the United States, KS primarily occurs among people infected with human immunodeficiency virus (HIV). The burden of HIV is disproportionately high among blacks, which may in part reflect less access to advanced antiretroviral therapies as well as higher rates of transmission among men who have sex with men.²² In 2013, the rate

of new HIV cases among blacks was 55.9 per 100,000 population compared with 6.6 per 100,000 population among whites.²³ Higher rates of stomach cancer in blacks are limited to noncardia gastric cancers. This disparity may reflect higher rates of *Helicobacter pylori* infection among blacks, which is the most important risk factor for noncardia tumors.²⁴ High consumption of salt and grilled meat also increase risk for this type of stomach cancer.²⁵ The reasons for higher rates of myeloma among blacks are not currently known.²⁶

Incidence rates for all cancers combined increased in blacks from the mid-1970s to the early 1990s; with a steeper slope in males than in females (Fig. 2). However, during the most recent time period (2003–2012), incidence rates decreased by 2.0% per year in black males but were stable in females, similar to the pattern in whites (Table 3). The declines in men are largely driven by cancers of the lung and prostate.

Mortality

About 69,410 blacks are expected to die from cancer in 2016, including 35,660 men and 33,750 women. Cancer is the second leading cause of death in blacks, accounting for 23% of all deaths in 2012 (Table 4). Lung cancer accounts for the largest number of cancer deaths among both men (27%) and women (22%), followed by prostate cancer in men (12%), and breast cancer in women (19%) (Fig. 1). For both men and women, colorectal cancer (CRC) is expected to be the third leading cause of cancer death.

TABLE 3. Fixed-Interval Trends (Annual Percent Change) in Cancer Incidence and Death Rates, 2003 to 2012

		MALE		FEMALE	
		BLACK	WHITE	BLACK	WHITE
All sites	Incidence	−2.0*	−1.2*	0.1	0.1
	Death	−2.5*	−1.6*	−1.5*	−1.3*
Lung & bronchus	Incidence	−2.5*	−2.3*	−1.1*	−1.0*
	Death	−3.3*	−2.5*	−1.6*	−1.2*
Prostate	Incidence	−3.4*	−4.2*	—	
	Death	−3.6*	−3.4*	—	
Female breast	Incidence	—		0.3*	−0.1
	Death	—		−1.4*	−1.8*
Colorectum	Incidence	−3.0*	−3.3*	−3.1*	−2.9*
	Death	−2.5*	−3.0*	−3.3*	−2.9*
Uterine cervix	Incidence	—		−3.8*	−2.6*
	Death	—		−2.6*	−0.9*

*The annual percent change from 2003 to 2012 is significantly different from zero.

Sources: *Incidence*: Surveillance, Epidemiology, and End Results (SEER) Program, SEER 13 registries, National Cancer Institute.⁵ *Mortality*: National Center for Health Statistics, Centers for Disease Control and Prevention.¹² Death rates for blacks and whites exclude those of Hispanic ethnicity.

TABLE 4. Leading Causes of Death Among Non-Hispanic (NH) Blacks and Whites, 2012

ALL AGES								
CAUSE OF DEATH	NH BLACK				NH WHITE			
	RANK	NUMBER	PERCENT OF TOTAL DEATHS	DEATH RATE*	RANK	NUMBER	PERCENT OF TOTAL DEATHS	DEATH RATE*
Heart diseases	1	69,139	24%	216.8	1	481,976	24%	171.2
Cancer	2	66,560	23%	199.2	2	462,493	23%	170.2
Cerebrovascular diseases	3	15,712	5%	50.7	4	100,152	5%	35.5
Diabetes	4	12,835	4%	39.8	7	50,442	3%	18.5
Accidents (unintentional injuries)	5	12,447	4%	32.5	5	99,284	5%	43.7
All causes		291,148		889.0		2,016,830		742.3

CHILDREN AGES 1-14								
CAUSE OF DEATH	NH BLACK				NH WHITE			
	RANK	NUMBER	PERCENT OF TOTAL DEATHS	DEATH RATE*	RANK	NUMBER	PERCENT OF TOTAL DEATHS	DEATH RATE*
Accidents	1	617	29%	7.1	1	1,543	32%	5.0
Homicide	2	233	11%	2.7	4	279	6%	0.9
Cancer	3	204	10%	2.4	2	703	15%	2.3
Congenital anomalies (birth defects)	4	181	8%	2.1	3	395	8%	1.3
Chronic lower respiratory diseases	5	104	5%	1.2	11	36	1%	0.1
All causes		2,144		24.6		4,761		15.4

*Rates are per 100,000 and age adjusted to the 2000 US standard population.

Source: National Center for Health Statistics, Centers for Disease Control and Prevention.¹²

Death rates are higher in blacks than in whites for most cancers (Table 5).

Death rates among blacks for all cancers combined have been decreasing since the early 1990s, with larger declines in men than in women (Fig. 2). The reduction in overall cancer death rates since 1990 in men and since 1991 in women translates to the avoidance of more than 300,000 deaths from cancer (Fig. 3). In fact, since the mid-1990s, death rates have declined faster among blacks than among whites (Table 3). As a result, the overall racial gap is narrowing, particularly among men. In 1990, the cancer death rate in males was 47% higher in blacks than in whites, but was 24% higher in 2012. Among females, the disparity decreased from 19% higher in 1991 to 14% higher in 2012.

Notably, the higher death rate in black women compared with white women occurs despite their lower incidence rate. Higher death rates in blacks are due largely to cancers of the breast and colorectum in women and to cancers of the prostate, lung and bronchus, and colorectum in men. However, in recent years, death rates for lung and prostate cancer have decreased faster in black men than in white men, contributing to the recent narrowing of the overall racial disparity (Fig. 4). In addition, lung (male and female) and cervical cancer death rates have converged for young

blacks and whites.^{27,28} In contrast, the racial disparity continues to widen for breast cancer in women and remains level for CRC in men (Fig. 4): cancers for which incidence and mortality are largely influenced by access to screening and treatment.

Variations in cancer incidence and death rates for selected cancers by state are presented in Tables 6 and 7. There is wide variation in rates by state, particularly for cancers closely tied to behavioral factors like smoking. For example, the lung cancer incidence rate in black men in Kentucky (136 per 100,000) is double that in Colorado (64 per 100,000) due to historic differences in smoking prevalence.

Survival and stage distribution

The 5-year relative survival rate is lower in blacks than in whites for every stage of diagnosis for most cancer sites (Fig. 5). Much of the difference in survival is believed to be due to barriers that limit access to timely, appropriate, and high-quality medical care.²⁹⁻³³ Furthermore, blacks are more likely to be diagnosed at later stages of the disease for most cancer sites (Fig. 6) when treatment choices are more limited and often less effective. It is recognized that these issues largely reflect socioeconomic disparities. Some studies suggest that blacks who receive cancer treatment and medical care similar to

TABLE 5. Comparison of Cancer Death Rates Between Non-Hispanic (NH) Blacks and Whites, United States, 2008–2012

MALE					FEMALE				
CANCER	NH BLACK RATE*	NH WHITE RATE*	ABSOLUTE DIFFERENCE†	RATE RATIO‡	CANCER	NH BLACK RATE*	NH WHITE RATE*	ABSOLUTE DIFFERENCE†	RATE RATIO‡
Stomach	9.4	3.6	5.8	2.58§	Stomach	4.5	1.8	2.7	2.48§
Prostate	47.2	19.9	27.3	2.38§	Myeloma	5.4	2.4	3.0	2.22§
Larynx	3.7	1.8	1.9	2.02§	Uterine cervix	4.1	2.0	2.1	2.00§
Myeloma	7.8	4.0	3.8	1.95§	Uterine corpus	7.8	4.1	3.7	1.92§
Liver & IHB	12.8	7.6	5.2	1.69§	Liver & IHB	4.4	3.1	1.3	1.43§
Colon & rectum	27.6	18.2	9.4	1.52§	Breast	31.0	21.9	9.1	1.42§
Oral cavity & pharynx	5.2	3.8	1.4	1.36§	Colon & rectum	18.2	12.9	5.3	1.41§
Pancreas	15.4	12.7	2.7	1.21§	Pancreas	12.6	9.5	3.1	1.32§
Lung & bronchus	74.9	62.2	12.7	1.20§	Esophagus	2.0	1.6	0.4	1.28§
Kidney & renal pelvis	5.7	5.9	−0.2	0.97	Urinary bladder	2.6	2.3	0.3	1.12§
Hodgkin lymphoma	0.4	0.5	−0.1	0.94	Kidney & renal pelvis	2.6	2.6	0.0	1.02
Esophagus	7.1	8.0	−0.9	0.89§	Lung & bronchus	36.7	41.1	−4.4	0.89§
Leukemia	8.1	9.9	−1.8	0.82§	Leukemia	4.8	5.4	−0.6	0.89§
Non-Hodgkin lymphoma	5.9	8.3	−2.4	0.71§	Hodgkin lymphoma	0.3	0.3	0.0	0.89
Urinary bladder	5.4	8.4	−3.0	0.65§	Ovary	6.8	8.2	−1.4	0.83§
Brain & ONS	3.2	6.0	−2.8	0.53§	Non-Hodgkin lymphoma	3.6	5.0	−1.4	0.71§
Melanoma of the skin	0.5	5.0	−4.5	0.09§	Brain & ONS	2.2	3.9	−1.7	0.55§
					Melanoma of the skin	0.4	2.1	−1.7	0.18§
All sites	267.7	210.6	57.1	1.27§	All sites	170.4	149.2	21.2	1.14§

IHB indicates intrahepatic bile duct; ONS, other nervous system.

*Rates are per 100,000 and age adjusted to the 2000 US standard population.

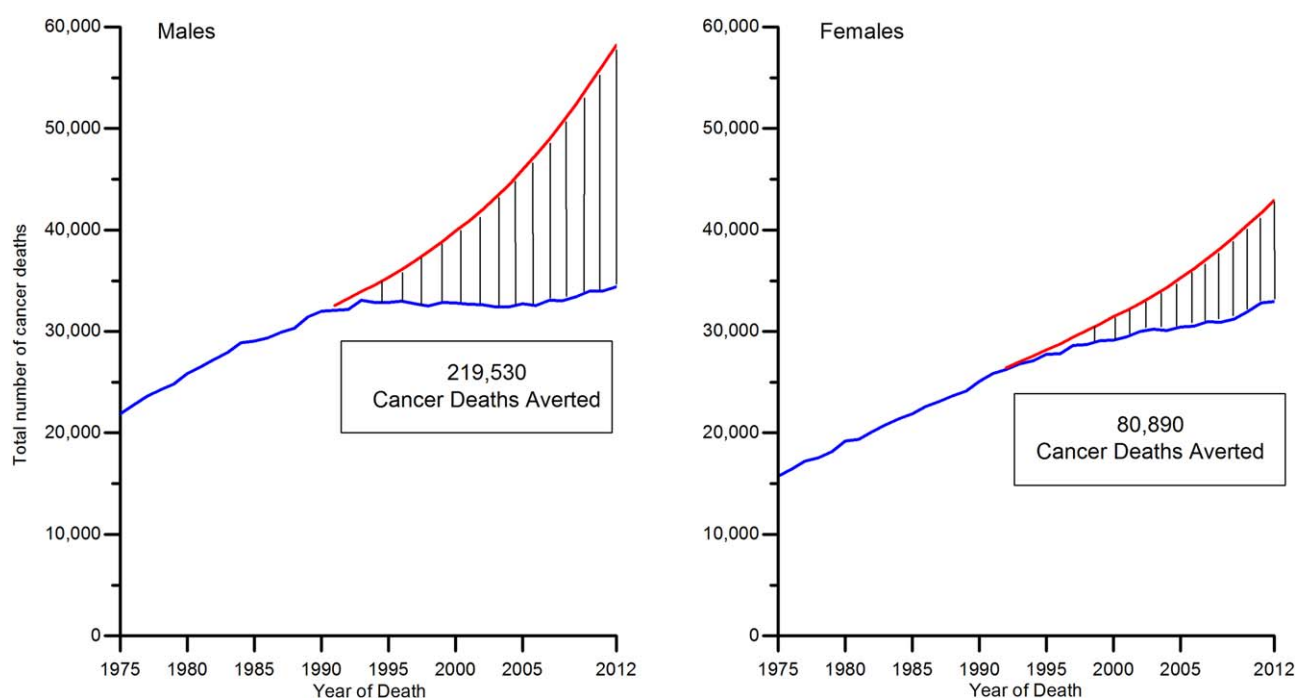
†The absolute difference is the rate in blacks minus the rate in whites.

‡The rate ratio is the unrounded rate in blacks divided by the unrounded rate in whites.

§The rate ratio is significantly different from one ($P < .05$).

Note: Sites are listed in descending order by rate ratio.

Source: National Center for Health Statistics, Centers for Disease Control and Prevention.¹²

**FIGURE 3. Total Number of Cancer Deaths Averted From 1991 to 2012 in Black Men and From 1992 to 2012 in Black Women.**

The blue line represents the actual number of cancer deaths recorded in each year, and the red line represents the expected number of cancer deaths if cancer death rates had remained at peak rates.

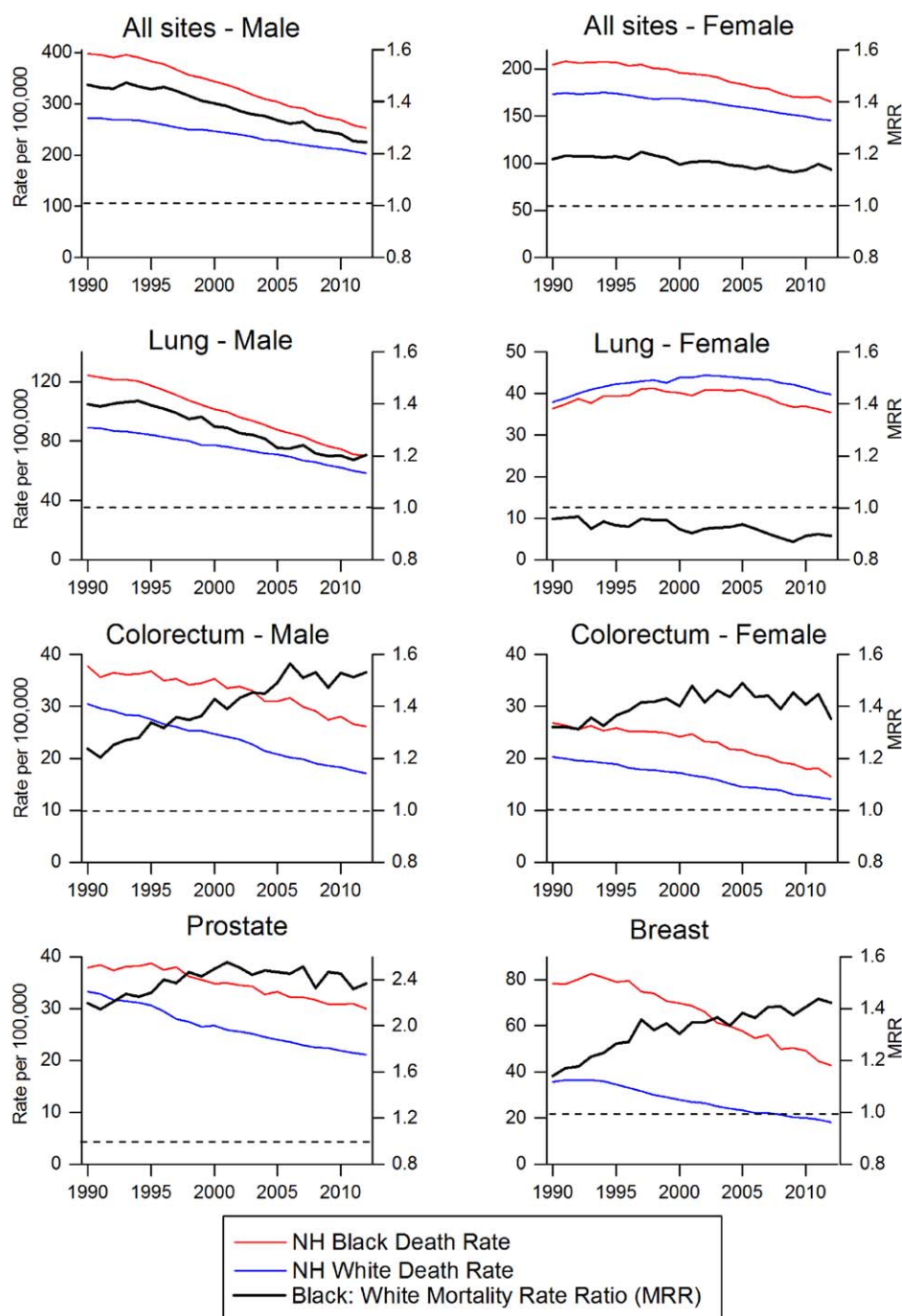


FIGURE 4. Trends in Cancer Death Rates and Mortality Rate Ratios by Site and Race/Ethnicity, United States, 1990 to 2012.

Note: Vertical scales (death rates and rate ratios) differ by site. NH indicates non-Hispanic.

Source: National Center for Health Statistics, Centers for Disease Control and Prevention.¹²

that of whites experience similar outcomes.^{31,34} However, other studies report that racial disparities persist even after controlling for socioeconomic factors and access to care.³⁵⁻³⁹ Higher rates of comorbid health conditions (eg, obesity, diabetes, and hypertension) among black patients can affect the delivery of optimal treatment and are also thought to contribute to differences in survival.⁴⁰⁻⁴² Although there is limited evidence that differing responses to cancer therapy contribute to racial disparities in survival,

blacks and other racial minorities are underrepresented in clinical trials, which makes it more difficult to assess the efficacy of cancer therapies in these groups.^{43,44}

The overall 5-year relative survival rate among blacks has improved from approximately 27% during 1960 through 1963 to 62% during 2005 through 2011.⁸ Survival during the corresponding period in whites increased from 39% to 70%. Increases in survival over time reflect earlier diagnoses and improvements in treatment;

TABLE 6. Incidence Rates* for Selected Cancers in Non-Hispanic Black Males and Females by State, 2008–2012

	ALL CANCERS		LUNG AND BRONCHUS		COLON AND RECTUM		PROSTATE	BREAST	UTERINE CERVIX
	MALE	FEMALE	MALE	FEMALE	MALE	FEMALE	MALE	FEMALE	FEMALE
Alabama	602.3	379.8	103.6	38.2	65.9	46.5	220.9	125.9	10.3
Alaska	564.8	360.8	91.6	†	†	†	201.0	141.7	†
Arizona	423.1	347.9	65.9	51.4	45.2	37.3	132.0	103.2	8.2
Arkansas‡§	597.8	357.5	115.4	49.1	62.2	48.9	205.1	101.3	11.4
California	572.3	417.5	79.7	51.7	61.3	47.3	197.2	129.1	8.2
Colorado	502.1	374.2	64.3	47.1	48.9	36.7	187.9	120.2	5.9
Connecticut	592.8	408.8	77.7	47.9	62.0	42.5	217.2	122.1	7.5
Delaware	612.3	414.9	87.5	53.4	47.9	36.8	231.5	127.5	8.9
District of Columbia	654.2	453.3	99.4	57.4	63.2	48.4	215.9	137.9	11.3
Florida	533.2	365.9	75.1	36.9	54.4	39.1	197.0	109.7	12.0
Georgia	609.0	392.0	91.7	42.4	60.3	44.5	235.3	124.1	9.2
Hawaii	473.6	360.2	†	†	39.6	†	184.2	134.0	†
Idaho	468.7	366.6	†	†	†	†	†	†	†
Illinois	622.5	437.7	102.1	65.3	71.3	50.5	211.7	126.8	12.7
Indiana	544.0	421.3	109.5	63.6	58.8	47.5	150.2	123.7	8.8
Iowa	586.9	453.6	99.8	86.4	58.5	51.2	164.6	111.6	†
Kansas	621.0	450.7	104.0	64.8	62.8	51.5	214.5	131.4	†
Kentucky	636.8	458.4	135.6	81.6	65.9	52.5	170.3	133.2	7.4
Louisiana	655.9	422.6	113.1	52.0	72.6	51.8	223.4	130.0	12.0
Maine	425.8	272.8	†	†	†	†	161.9	†	†
Maryland	547.1	402.3	76.4	49.3	50.7	39.6	203.1	130.2	8.3
Massachusetts	568.2	387.1	72.9	41.5	49.3	37.0	218.9	115.1	8.3
Michigan	636.1	429.3	104.0	63.1	59.4	44.5	223.1	122.7	8.8
Minnesota‡	-	-	-	-	-	-	-	-	-
Mississippi	648.8	408.5	116.2	47.7	75.8	55.8	225.3	124.0	12.5
Missouri	582.1	450.2	106.5	70.8	67.2	49.1	171.5	135.6	10.5
Montana	489.9	†	†	†	†	†	†	†	†
Nebraska	594.6	456.2	107.6	61.0	80.8	57.7	177.9	134.2	†
Nevada‡¶	468.9	379.9	69.6	48.0	57.8	47.4	141.7	116.8	9.9
New Hampshire	497.5	288.9	†	†	†	†	190.9	†	†
New Jersey	606.7	415.0	80.6	50.4	59.7	44.9	234.1	124.4	11.2
New Mexico	405.6	320.6	81.8	32.7	28.5	31.5	141.4	107.2	†
New York	613.1	406.2	74.6	45.0	57.5	41.2	247.3	119.2	11.4
North Carolina	609.1	403.2	103.6	47.2	58.4	41.2	213.7	128.1	8.7
North Dakota	541.0	†	†	†	†	†	†	†	†
Ohio	572.1	405.7	102.8	64.2	56.1	38.9	190.7	121.0	7.7
Oklahoma	613.2	421.0	104.8	56.4	55.3	46.4	229.1	131.3	9.4
Oregon	531.3	392.2	87.8	59.1	63.4	39.4	174.3	121.6	†
Pennsylvania	624.6	467.1	104.3	71.8	60.9	43.6	198.7	131.1	11.3
Rhode Island	505.3	380.6	67.6	66.2	31.9	33.1	170.0	104.8	†
South Carolina	583.8	388.0	94.4	41.0	57.1	40.6	207.2	125.1	9.3
South Dakota	315.2	309.8	†	†	†	†	†	†	†
Tennessee	602.8	407.3	109.6	53.1	64.0	45.9	203.8	126.2	10.9
Texas	572.4	406.8	99.9	52.3	63.9	45.1	175.8	120.3	11.1
Utah	551.7	386.1	†	†	†	†	210.9	108.6	†
Vermont	325.9	†	†	†	†	†	†	†	†
Virginia	578.8	392.8	95.1	49.5	54.2	41.0	208.9	129.8	7.6
Washington	573.1	417.1	80.9	55.8	47.4	36.0	196.2	127.3	6.9
West Virginia	565.2	369.3	99.5	47.1	65.6	38.2	192.2	115.5	†
Wisconsin	684.9	458.7	127.8	74.1	69.4	44.1	217.8	126.0	11.3
Wyoming	264.2	236.7	†	†	†	†	†	†	†
United States	592.3	408.1	93.4	51.4	60.3	44.1	208.7	124.3	10.0

*Rates are per 100,000 and age adjusted to the 2000 US standard population.

†Rates are suppressed when they are based on fewer than 25 cases.

‡This state's data are not included in US combined rates.

§Rates are based on incidence data for 2008–2009.

||Incidence data not submitted to North American Association of Central Cancer Registries.

¶Rates are based on incidence data for 2008–2010.

Source: North American Association of Central Cancer Registries.⁹

however, not all persons have benefited equally from these advances. Importantly, improvements in survival do not always indicate progress against cancer, such as

when they result from the detection of indolent cancers (overdiagnosis) or when early diagnosis does not extend lifespan (lead time bias).

TABLE 7. Death Rates* for Selected Cancers in Non-Hispanic Black Males and Females by State, 2008-2012

	ALL CANCERS		LUNG AND BRONCHUS		COLON AND RECTUM		PROSTATE	BREAST	UTERINE CERVIX
	MALE	FEMALE	MALE	FEMALE	MALE	FEMALE	MALE	FEMALE	FEMALE
Alabama	298.3	167.5	87.3	29.8	31.2	19.7	56.7	30.7	4.7
Alaska	247.4	128.8	†	†	†	†	†	†	†
Arizona	204.9	152.8	47.7	32.5	24.3	17.8	32.1	29.7	†
Arkansas	304.6	183.2	99.3	38.7	32.3	21.7	53.0	31.4	6.4
California	250.3	181.4	63.4	38.8	26.3	19.9	48.4	33.1	3.6
Colorado	219.9	154.5	54.6	35.5	22.8	15.0	51.2	26.0	†
Connecticut	224.6	148.2	55.7	31.0	19.1	13.5	39.7	24.6	†
Delaware	248.5	168.6	68.9	37.4	16.5	15.3	39.8	26.5	†
District of Columbia	291.7	190.3	74.2	41.0	26.2	20.3	48.0	34.0	4.0
Florida	232.0	147.2	59.8	25.1	23.5	16.4	46.2	28.5	4.9
Georgia	264.7	155.6	71.0	29.5	27.1	17.6	52.6	29.5	3.9
Hawaii	194.6	125.3	†	†	†	†	†	†	†
Idaho	†	†	†	†	†	†	†	†	†
Illinois	284.6	188.4	80.8	45.6	30.8	20.5	48.8	32.8	5.5
Indiana	282.1	190.9	90.6	48.1	26.8	19.1	43.9	31.0	3.5
Iowa	291.9	190.2	71.6	52.6	26.4	25.7	48.7	25.6	†
Kansas	270.5	197.9	80.1	54.9	27.1	22.1	44.3	29.4	†
Kentucky	292.9	191.4	102.1	56.8	26.2	20.7	40.4	32.7	3.2
Louisiana	303.5	182.0	94.0	40.0	31.3	19.6	45.2	34.8	4.5
Maine	†	†	†	†	†	†	†	†	†
Maryland	250.5	166.0	66.2	36.7	26.6	17.4	43.4	30.6	3.3
Massachusetts	223.4	146.9	51.1	26.9	19.7	14.0	42.5	23.7	2.5
Michigan	271.7	183.1	82.0	45.1	28.0	17.4	38.8	33.1	3.8
Minnesota	226.2	151.0	53.0	30.4	15.4	9.3	33.6	21.7	†
Mississippi	323.1	175.7	98.4	35.1	33.8	22.2	59.6	33.3	5.6
Missouri	272.8	188.2	79.6	47.1	28.8	18.6	41.4	33.7	4.5
Montana	†	†	†	†	†	†	†	†	†
Nebraska	287.6	186.8	86.2	48.5	40.4	19.2	38.7	29.0	†
Nevada	218.6	154.4	59.2	39.5	23.6	19.6	37.4	29.3	†
New Hampshire	146.1	†	†	†	†	†	†	†	†
New Jersey	263.2	170.9	65.7	34.9	30.2	18.8	49.4	32.5	4.1
New Mexico	218.6	157.2	73.6	†	†	†	†	30.2	†
New York	226.7	154.4	54.6	30.0	23.8	15.6	44.8	28.4	4.4
North Carolina	282.7	160.3	83.2	32.7	27.2	17.2	52.3	28.8	3.3
North Dakota	†	†	†	†	†	†	†	†	†
Ohio	278.4	181.8	86.8	47.8	28.0	16.5	44.6	30.9	3.6
Oklahoma	287.4	188.5	84.5	40.6	33.2	22.5	55.9	35.4	4.8
Oregon	261.3	170.4	69.5	48.7	27.3	†	47.9	28.5	†
Pennsylvania	288.4	193.3	80.9	49.1	29.7	18.1	52.6	33.1	4.1
Rhode Island	203.2	127.3	47.5	36.9	†	†	†	28.1	†
South Carolina	286.2	162.6	75.6	30.0	28.0	17.5	52.8	29.2	4.1
South Dakota	†	†	†	†	†	†	†	†	†
Tennessee	321.3	186.3	97.3	41.0	34.8	21.4	53.0	33.9	5.2
Texas	274.6	177.4	81.4	38.5	30.5	19.9	39.5	33.7	4.2
Utah	162.6	148.8	†	†	†	†	†	†	†
Vermont	†	†	†	†	†	†	†	†	†
Virginia	271.1	165.3	76.1	35.9	26.9	17.1	47.3	31.7	2.9
Washington	241.2	157.8	61.0	36.8	20.6	13.3	48.4	25.0	†
West Virginia	282.7	173.2	89.8	40.2	35.2	17.5	56.0	26.6	†
Wisconsin	307.8	195.0	102.7	50.1	28.0	18.2	42.6	32.1	3.8
Wyoming	†	†	†	†	†	†	†	†	†
United States	267.7	170.4	74.9	36.7	27.6	18.2	47.2	31.0	4.1

*Rates are per 100,000 and age adjusted to the 2000 US standard population.

†Rates are suppressed when they are based on fewer than 25 deaths.

Source: National Center for Health Statistics, Centers for Disease Control and Prevention.¹²

Selected Cancer Sites

Female breast

Breast cancer is the most common cancer among black women and is the second leading cause of cancer death, with an estimated 30,700 new cases of breast cancer and

6310 deaths expected to occur in 2016. During 2008 through 2012, the overall breast cancer incidence rate in black women was 124.3 cases per 100,000 women, which was 3% lower than that in white women (128.1 per 100,000 women) (Table 2). However, rates were higher in

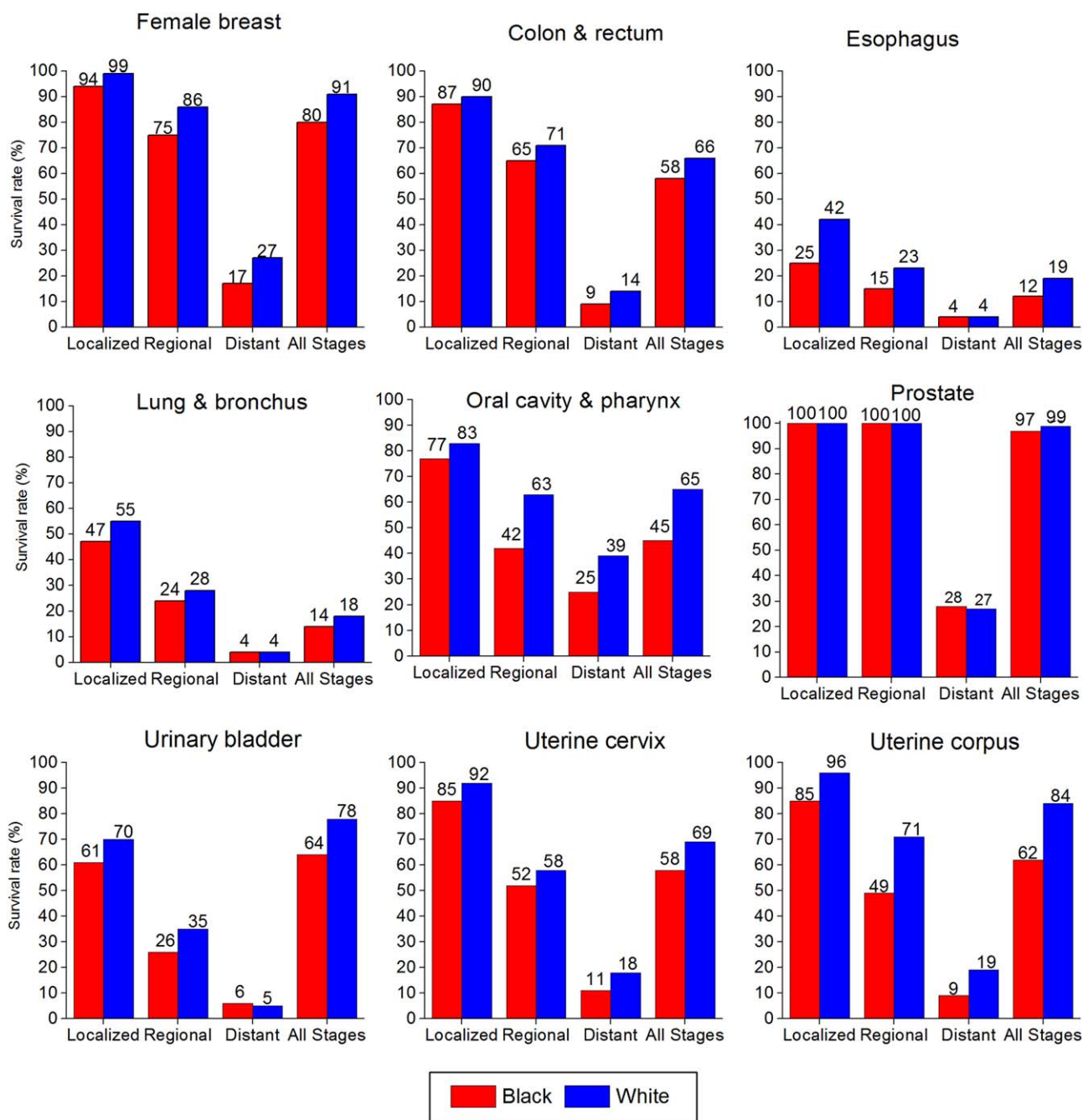


FIGURE 5. Five-year Relative Survival Rates for Selected Cancers by Race and Stage, United States, 2005 to 2011.

Survival rates are based on patients who were diagnosed between 2005 and 2011 and followed through 2012.
Source: Surveillance, Epidemiology, and End Results (SEER) Program, SEER 18 registries, National Cancer Institute.⁶

black women than in white women in 7 states (Alabama, Kentucky, Louisiana, Mississippi, Missouri, Oklahoma, and Tennessee) and were not significantly different in 24 states.⁴⁵ Breast cancer incidence rates are also higher among blacks than whites for women under age 45. The median age of diagnosis is 58 years for black women, compared with 62 years for white women.⁸ One in 9 black women is expected to be diagnosed with breast cancer in her lifetime, compared with 1 in 8 white women (Table 1).

Long-term breast cancer incidence trends are shown in Figure 7. Similar to the pattern among white women, rates among black women increased rapidly during the 1980s, largely because of increased detection due to mammography screening uptake. However, rates stabilized in white women (since 2004) but continued to increase in black women from 1986 to 2012 by 0.5% per year.⁸ As a result, incidence rates in black and white women converged in 2012. The continued increase in breast cancer

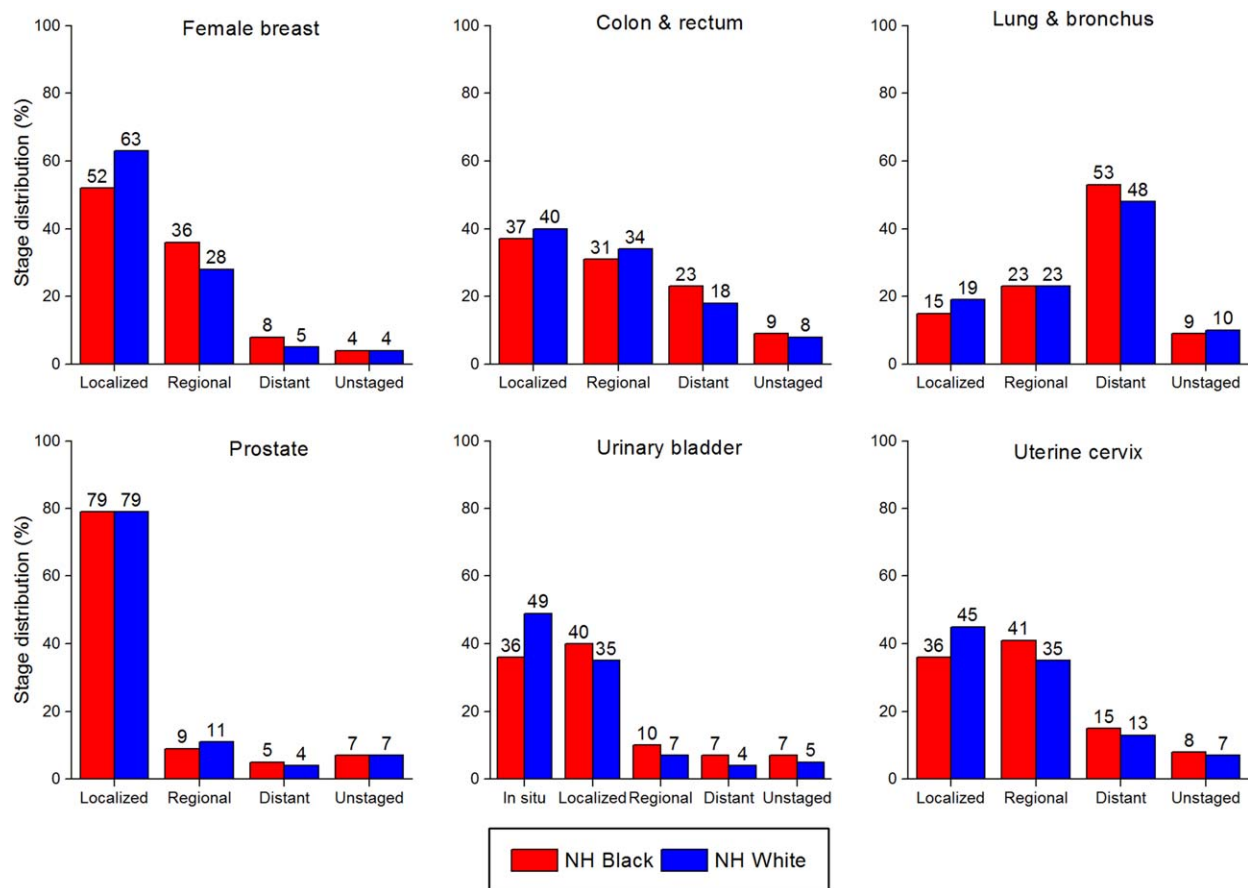


FIGURE 6. Stage Distribution for Selected Cancers in Non-Hispanic (NH) Blacks and Whites, United States, 2005 to 2011.

Percentages may not total 100% because of rounding.

Source: North American Association of Central Cancer Registries.⁹

incidence rates in black women may in part reflect the obesity epidemic. The prevalence of obesity (body mass index ≥ 30 kg/m²) increased from 38% during 1988 through 1994 to 57% in 2013 and 2014 for black women and from 23% to 38% over the same period for white women (Fig. 8).

Prior to the mid-1980s, breast cancer death rates for white and black women were similar. However, a larger increase in black than in white women from the mid-1970s to the early 1990s, followed by a slower decline in black women, has resulted in a widening disparity. Since 1990, breast cancer death rates dropped 23% in black women and 37% in white women (Fig. 4). As a result, breast cancer death rates in the most recent time period (2008-2012) are 42% higher in black women compared with white women, despite historically lower incidence rates. Higher death rates among black women likely reflect a combination of factors, including differences in stage at diagnosis, obesity, comorbidities, and tumor characteristics as well as access, adherence, and response to high-quality cancer treatment.⁴⁶⁻⁵²

The 5-year relative survival rate for breast cancer diagnosed in 2005 through 2011 among black women was 80%, compared with 91% among white women (Fig. 5).

This difference can be attributed to both later stage at detection and poorer stage-specific survival among black women. Despite similar reported mammography screening rates (Table 8), only about half (52%) of breast cancers are diagnosed at a local stage among black women, compared with 63% among white women (Fig. 6).

Later stage at diagnosis among black women has been largely attributed to lower frequency of and longer intervals between mammograms and lack of timely follow-up of abnormal results.⁵³⁻⁵⁵ Lower stage-specific survival has been explained in part by unequal access to and receipt of prompt, high-quality treatment among black women compared with white women.^{48,56-59} Aggressive tumor characteristics are more common in breast cancers diagnosed in black women than in other racial/ethnic groups.^{45,60-62} For example, 22% of breast cancers in black women are triple negative (estrogen receptor-negative, progesterone receptor-negative, and human epidermal growth factor receptor 2-negative) compared with 10% to 12% of those among women of other races/ethnicities in the United States.⁴⁵ These proportions are even higher among premenopausal black breast cancer patients.⁶³ Triple-negative breast cancers, which include the basal-type subset

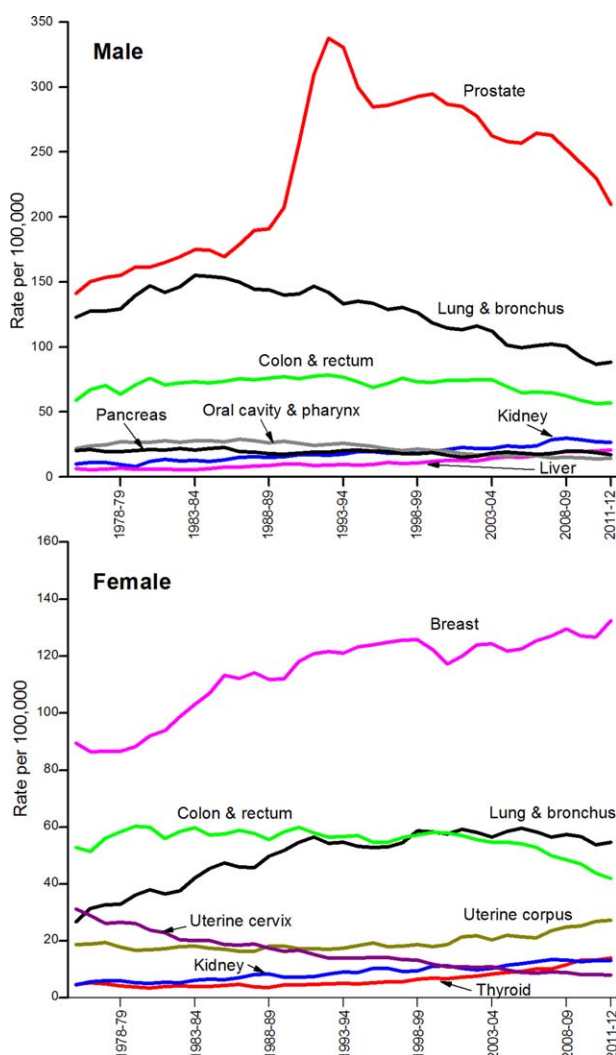


FIGURE 7. Trends in Cancer Incidence Rates Among Blacks, United States, 1975 to 2012.

Rates are delay adjusted and age adjusted to the 2000 US standard population and are 2-year moving averages.
Source: SEER Program, SEER 9 registries, National Cancer Institute.⁴

of breast cancers, are more aggressive and have a poorer prognosis, in part because there are currently no targeted therapies for these tumors.^{64,65} Some studies suggest that black women are more likely to be diagnosed with triple-negative and basal-like breast cancers as a result of shared African ancestry.^{60,66} A recent analysis from the African American Breast Cancer Epidemiology and Risk (AMBER) Consortium identified eight genes that may be involved in the etiology of breast cancer in black women, including three linked to estrogen receptor-negative disease.⁶⁷ Other studies have found that risk factors vary for breast cancer subtypes and that obesity and certain reproductive patterns (including multiparity, early age at first pregnancy, and lower rates of breastfeeding) which are more common in black women, are linked to increased risk for aggressive breast cancer subtypes.^{63,68-72}

Colon and rectum

CRC is the third most common cancer diagnosed among both black men and women and the second most common for both sexes combined, with 17,240 new cases expected to be diagnosed in 2016 (Fig. 1). The median age of diagnosis for CRC is 66 years for black men and 70 years for black women, compared with ages 72 and 77 years for white men and women, respectively.⁸ CRC is also the third leading cause of cancer death in black men and women, with 7030 CRC deaths expected in 2016.

Incidence rates for CRC are 27% higher in black men and 22% higher in black women compared with white men and women, respectively (Table 2). Blacks have higher rates of obesity and higher rates of physical inactivity (Table 9), which are known risk factors for CRC. Results from the National Institutes of Health-AARP Diet and Health Study suggest that health behaviors (diet, physical activity, and smoking) and body mass index explain more than a third (36%) of the increased risk for CRC associated with low socioeconomic status, for which blacks are disproportionately represented.⁷³ In addition, uptake of CRC screening, which can prevent cancer through the removal of precancerous polyps as well as detect cancer at an early stage, has been slower in blacks than in whites and remains slightly lower. From 2000 to 2013, CRC screening increased from 32% to 59% in blacks and from 40% to 61% in whites.²⁰ Before 1989, incidence rates were

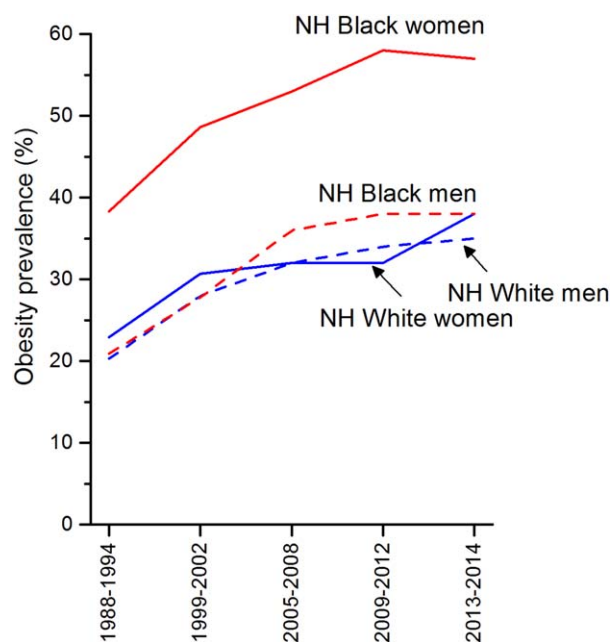


FIGURE 8. Trends in Adult Obesity (Body Mass Index ≥ 30 kg/m²) Prevalence (%) by Sex and Race/Ethnicity, United States, 1988 to 2014.

NH indicates non-Hispanic.

Sources: 1988-2012: Health, United States, 2014: With Special Feature on Adults Ages 55-64. 2013-2014: Centers for Disease Control and Prevention. National Health and Nutrition Examination Survey, 2014. Public use data file.²¹

TABLE 8. Human Papillomavirus Vaccination (2014) and Use of Cancer Screening Examinations and Tests (2013), United States

	NH BLACK (%)	NH WHITE (%)
HPV vaccination* (youth 13-17 years)		
Girls		
≥ 1 dose	66	56
≥ 3 doses	39	38
Boys		
≥ 1 dose	42	36
≥ 3 doses	20	19
Breast cancer (women 40 years and older)		
Mammogram in the past 2 years	66	66
Cervical cancer (women ages 21-65 years)†		
Pap test in the past 3 years	82	83
Colorectal cancer (adults 50 years and older)		
FOBT in the past year	9	7
Endoscopy‡	57	58
FOBT or endoscopy§	59	61
Prostate cancer (men 50 years and older)		
PSA test in the past year	33	37

NH indicates non-Hispanic; Pap, Papanicolaou; FOBT, fecal occult blood test; PSA, prostate-specific antigen.

*Complete vaccination series consists of 3 doses.

†Among women with intact uteri.

‡Sigmoidoscopy in the past 5 years or colonoscopy in the past 10 years.

§FOBT in the past year, sigmoidoscopy in the past 5 years, or colonoscopy in the past 10 years.

Note: Estimates for screening are age-adjusted to the 2000 US standard population.

Sources: *Vaccination*: National Immunization Survey-Teen, 2014.¹¹⁸ *Screening*: Centers for Disease Control and Prevention. National Health Interview Survey, 2013. Public use data file.²⁰

predominantly higher in white men than in black men and were similar for women of both races. Since 1989, however, incidence rates have been higher for blacks than for whites in both men and women. This crossover likely reflects racial differences in risk factor trends and/or greater access to and utilization of recommended screening tests by whites.⁷⁴ From 2003 to 2012, incidence rates decreased by 3.0% per year in black men and by 3.1% per year in black women, similar to declines in whites (Table 3).

The racial disparity is more striking for CRC death rates: rates are 52% higher in black men and 41% higher in black women (Table 5). One model-based study estimated that 19% of the racial disparity in CRC mortality rates can be attributed to lower screening rates and 36% can be attributed to lower stage-specific survival among blacks.⁷⁵ Similar to the pattern for incidence rates, CRC mortality rates were historically higher in whites compared with blacks, with the crossover occurring around 1979 for women and 1984 for men. From 2003 through 2012, annual declines in mortality rates were higher in black women than white women (3.3% vs 2.9%) but were lower in black men than in white men (2.5% vs 3.0%) (Table 3). As a result, the racial gap appears to be shrinking in women, whereas rates in men have

remained about 50% higher in blacks than in whites since 2005 (Fig. 4). Smaller declines in death rates for distant-stage disease in blacks than in whites (5% vs 33%) appear to be driving the overall mortality differential.⁷⁶

The 5-year relative survival rate for CRC among blacks improved from 45% during 1975 through 1977 to 59% during 2005 through 2011; however, this improvement was smaller than that in whites (from 50% to 67% over the same period).⁸ Some of the racial disparity in survival is because of later stage at diagnosis—37% of CRCs in blacks are diagnosed at a localized stage compared with 40% in whites (Fig. 6). However, lower 5-year relative survival rates are also seen in black CRC patients within each stage at diagnosis (Fig. 5). Racial disparities in CRC survival largely reflect differences in treatment, socioeconomic status, and comorbidities.^{35,77-80} Numerous studies document that black CRC patients are less likely than white patients to receive recommended surgical treatment and adjuvant chemotherapy.⁸¹⁻⁸³ Notably, a recent study reported that, when black and white stage III CRC patients received similar adjuvant chemotherapy (combined folinic acid, fluorouracil, and oxaliplatin

TABLE 9. Risk Factors for Cancer by Sex and Race/Ethnicity, Adults, United States, 2013-2014

	NH BLACK (%)	NH WHITE (%)
Obesity (BMI ≥ 30.0)*		
All	48	36
Men	38	35
Women	57	38
Overweight (BMI 25.0-29.9)*		
All	28	33
Men	31	40
Women	25	26
No leisure-time physical activity†		
All	38	26
Men	34	25
Women	41	27
Met recommendations for aerobic activity†‡		
All	44	54
Men	51	56
Women	38	52
Current cigarette smoking†§		
All	18	19
Men	22	20
Women	14	18

NH indicates non-Hispanic; BMI, body mass index, kg/m².

*Among adults 20 years and older.

†Among adults 18 years and older.

‡Includes 150 minutes of moderate-intensity activity or 75 minutes of vigorous-intensity activity each week.

§Ever smoked 100 cigarettes in lifetime and smoking every day or some days at time of survey.

Note: Estimates are age-adjusted to the 2000 US standard population.

Sources: *BMI*: Centers for Disease Control and Prevention. National Health and Nutrition Examination Survey, 2013-2014. Public use data file.²¹ *Physical activity and smoking*: Centers for Disease Control and Prevention. National Health Interview Survey, 2014. Public use data file.²⁰

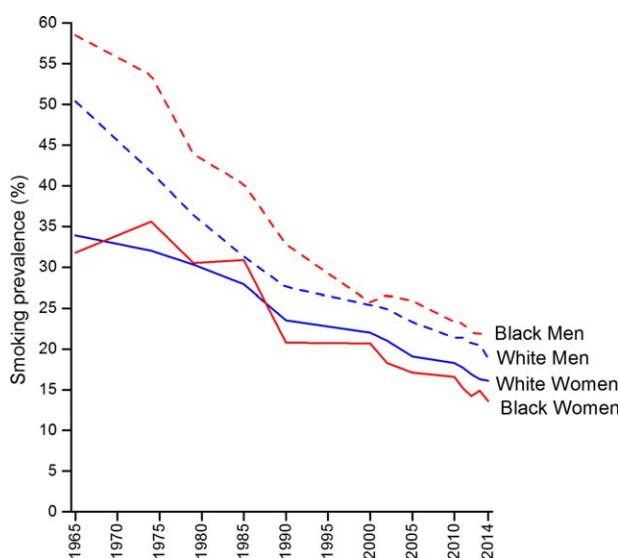


FIGURE 9. Trends in Adult Smoking Prevalence (%) by Sex and Race, 1965 to 2014.

Sources: 1965-2013: Health, United States, 2014: With Special Feature on Adults Aged 55-64. 2014: Centers for Disease Control and Prevention. National Health Interview Survey, 2014. Public use data file.²⁰

[FOLFOX]) in a phase 3 clinical trial, those aged 50 years and older had similar outcomes; however, younger blacks had shorter disease-free survival (hazard ratio, 1.80; 95% confidence interval, 1.21-2.66) and time to recurrence (hazard ratio, 1.77; 95% confidence interval, 1.18-2.65) compared with whites.⁸⁴

Lung and bronchus

Lung cancer is the second leading cause of cancer in black men and women, with 13,720 men and 11,010 women expected to be newly diagnosed in 2016 (Fig. 1). Lung cancer is also the leading cause of cancer death in blacks, with 9710 men and 7340 women expected to die from this disease in 2016. Black men have higher lung cancer rates than white men, but the reverse is true for women, reflecting race and sex differences in historic smoking patterns (Fig. 9). The lung cancer incidence rate is 18% higher in black men compared with white men; however, among women, the rate is 13% lower among blacks (Table 2). Lung cancer trends are similar in blacks and whites. In black men, lung cancer incidence rates increased rapidly until the mid-1980s, but have since been steadily declining (Fig. 7). In contrast, in black women, rates increased until the early 2000s and have subsequently begun to decline. From 2003 to 2012, lung cancer incidence rates decreased slightly faster in black men and women than in white men and women (Table 3).

The lung cancer mortality rate is higher for black men than for any other racial or ethnic group. After increasing for decades, lung cancer death rates in men began to decline in 1990, with acceleration in the decline beginning in 1994. Similar to the pattern for incidence trends, the decline in lung cancer death rates has been faster in black men and

women (3.3% per year and 1.6% per year from 2003-2012, respectively) compared with white men (2.5% per year) and women (1.2% per year) (Table 3). The disparity in lung cancer death rates between black and white men has been substantially reduced overall (from an excess of 40% in the early 1990s to 20% in 2012) (Fig. 4) and has been eliminated in adults younger than 40 years.²⁸ The declines in lung cancer death rates are the result of decreases in smoking prevalence over the previous 40 years, which have been more rapid in blacks than in whites (Fig. 9). Furthermore, black adolescents initiate smoking at a much lower rate than their white counterparts.⁸⁵ If black youths continue to have lower smoking prevalence as they age, racial differences in lung cancer death rates should be eliminated in the next 40 to 50 years.²⁸

The 5-year overall relative survival rate for lung cancer is lower in blacks than in whites; 14% versus 18%, respectively (Fig. 5). When lung cancer is detected at a local stage, the 5-year relative survival rate in blacks is 47%; however, only 15% of lung cancer cases in blacks are detected at this early stage, because symptoms generally do not appear until the disease is advanced. Studies have shown that even when lung cancer is diagnosed early, blacks are less likely than whites to receive curative-intent surgery, even after accounting for socioeconomic factors.⁸⁶⁻⁸⁸ Other studies have found that, among lung cancer patients treated at Veterans Affairs or US Military Health System facilities, racial disparities in lung cancer outcomes diminished, although differences in receipt of treatment remained.⁸⁹⁻⁹¹

Prostate

Prostate cancer is the most commonly diagnosed cancer among black men and the second leading cause of cancer death. In 2016, approximately 29,530 cases of prostate cancer will be newly diagnosed and 4450 prostate cancer deaths will occur among black men. The median age of diagnosis for prostate cancer is 63 years for black men compared with 66 years for white men.⁸ It is estimated that 1 in 6 black men will be diagnosed with prostate cancer in their lifetime compared with 1 in 8 white men (Table 1).

During 2008 through 2012, the average annual prostate cancer incidence rate was 208.7 cases per 100,000 black men, which was 70% higher than the rate in white men (Table 2). Similar to the pattern in white men, incidence rates in black men increased sharply between 1989 and 1992—reflecting increased use of the prostate-specific antigen (PSA) blood test for the detection of prostate cancer—but have generally decreased thereafter (Fig. 7). During 2003 through 2012, prostate cancer incidence rates declined on average by 3.4% per year in black men and 4.2% per year in white men (Table 3). The decrease in incidence was driven by declines in early stage disease, and a recent study noted that the sharpest drop occurred after the release of

the US Preventive Services Task Force draft recommendation against PSA screening for men of all ages in 2011.⁹²

The only well established risk factors for prostate cancer are age, race, and family history of the disease. Men who have a first-degree relative with a history of prostate cancer are 2 to 3 times more likely to be diagnosed with the disease than men without a family history.⁹³ Black men and Jamaican men of African descent have the highest prostate cancer incidence rates worldwide, which may reflect differences in inherited genetic susceptibility.⁹⁴⁻⁹⁷

Similarly, black men have the highest mortality rate for prostate cancer of any racial or ethnic group in the United States, 2.4 times higher than the rate in white men (Table 5), in part reflecting higher incidence rates among black men. After a long period of increase, prostate cancer death rates in black men peaked in 1993 and declined steadily thereafter. Rates have been declining since 1996, by 3.6% per year in blacks and by 3.4% in whites, resulting in a narrowing disparity (Fig. 4). The decrease in prostate cancer mortality has been attributed to improved surgical and radiologic treatment and dissemination of hormonal therapy for advanced-stage disease.⁹⁸⁻¹⁰¹ The contribution of PSA testing is not clear. Results from a US-based randomized trial indicated no reduction in prostate cancer mortality as a result of PSA testing, while two European trials showed a modest benefit.¹⁰²⁻¹⁰⁴ Studies also suggest that black men are less likely to receive surgical treatment than white men with similar disease characteristics.¹⁰⁵⁻¹⁰⁸ Furthermore, a recent analysis of SEER-Medicare data concluded that, among localized prostate cancer patients who underwent radical prostatectomy, blacks were more likely to experience treatment delays and postoperative complications and were less likely to receive lymph node dissection compared with whites, although prostate cancer-specific and all-cause mortality were similar in both groups.¹⁰⁵ The lack of a difference in outcome despite treatment differences suggests that some patients with early stage disease may be overtreated.¹⁰⁹

The overall 5-year relative survival rate for prostate cancer is 97% among blacks and 99% among whites (Fig. 5). Eighty-eight percent of prostate cancers in black men are diagnosed at a local or regional stage (Fig. 6), for which the 5-year relative survival rate approaches 100%. Five-year survival rates drop to 28% when the cancer is diagnosed at distant stage.

Uterine cervix

An estimated 2290 new cases of invasive cervical cancer and 750 deaths are expected to occur among black women in 2016. The incidence rate of cervical cancer is 41% higher in black women than in white women (Table 2). However, a recent study suggests that the racial disparity may be even wider after adjusting incidence rates to account for women who have had a hysterectomy and thus are not at risk for cervical cancer.¹¹⁰

Nevertheless, the racial disparity has narrowed substantially, as rates have dropped faster among black women than among white women in recent years (Table 3). Notably, among women under age 50 years, incidence rates of cervical cancer converged between black and white women in the mid-2000s.²⁷ The median age of cervical cancer diagnosis is 51 years for black women compared with 48 years for white women.⁸

The overall 5-year relative survival rate for cervical cancer among black women is 58%, compared with 69% among white women (Fig. 5), partly because black women are more likely than white women to be diagnosed with regional-stage or distant-stage disease (Fig. 6) despite similar screening rates (Table 8). Racial differences in stage at diagnosis may be because of differences in the quality of screening and follow-up after abnormal results.^{111,112} Lower socioeconomic status is also associated with lower screening rates, later stage at diagnosis, and poorer survival.¹¹³⁻¹¹⁵

Virtually all cervical cancers are caused by persistent human papillomavirus (HPV) infection, particularly HPV types 16 and 18.¹¹⁶ The US Food and Drug Administration has approved and three vaccines are recommended for the prevention of the most common HPV genotypes. Two of the vaccines provide protection against HPV types causing approximately 70% of cervical cancers, while the third provides protection against HPV types associated with approximately 90% of invasive cervical cancers.¹¹⁷ Data from the 2014 National Immunization Survey-Teen found that, although HPV vaccine initiation was higher among black girls (66%) than among white girls (56%), completion of the three-vaccine series was similar (blacks, 39%; whites, 38%) (Table 8).¹¹⁸ Vaccinated women need to continue to receive recommended cervical cancer screening, because these vaccines only target the most common strains of HPV, and they also do not provide protection for those women who are already infected with HPV.^{119,120}

Data Limitations

The projected numbers of new cancer cases and cancer deaths should be interpreted cautiously, because these estimates are model-based and may vary considerably from year to year for reasons other than changes in cancer occurrence, including changes in methodology. Therefore, we discourage the use of these estimates to track year-to-year changes in cancer occurrence and death. The preferred data sources used for tracking cancer trends are age-standardized or age-specific cancer death rates from the National Center for Health Statistics and cancer incidence rates from SEER or NAACCR, although these data are 2 to 4 years old by the time they become available. Nevertheless, the American Cancer Society projections of the number of new cancer cases and deaths provide a reasonably accurate estimate of the current cancer burden in the United States.

Conclusions

Substantial progress has been made over the last several decades to reduce the disproportionate burden of cancer in blacks in the United States. The black-white disparity in cancer death rates has narrowed for all cancers combined in men and women and for lung and prostate cancer in men.

However, the racial gap in death rates has widened for breast cancer in women and remained level for CRC in men, likely because of suboptimal screening and treatment for blacks. Accelerating progress in eliminating racial disparities requires equitable access to services for prevention, early detection, and high-quality treatment. ■

References

- DeNavas-Walt C, Proctor BD. US Census Bureau, Current Population Reports, P60-252, Income and Poverty in the United States: 2014. Washington, DC: US Government Printing Office; 2015.
- US Census Bureau Population Division. Annual Social and Economic Supplement to the Current Population Survey. Table A-2: Percent of People 25 Years and Over Who Have Completed High School or College, by Race, Hispanic Origin and Sex: Selected Years 1940 to 2014. Washington, DC: US Census Bureau; 2015.
- Siegel RL, Fedewa SA, Miller KD, et al. Cancer statistics for Hispanics/Latinos, 2015. *CA Cancer J Clin*. 2015;65:457-480.
- Surveillance, Epidemiology, and End Results (SEER) Program. SEER*Stat Database: Incidence-SEER 9 Regs Research Data with Delay-Adjustment, Malignant Only, Nov 2014 Sub (1975-2012) <Katrina/Rita Population Adjustment>-Linked To County Attributes-Total US, 1969-2013 Counties. Bethesda, MD: National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, Surveillance Systems Branch; 2015.
- Surveillance, Epidemiology, and End Results (SEER) Program. SEER*Stat Database: Incidence-SEER 13 Regs Research Data with Delay-Adjustment, Malignant Only, Nov 2014 Sub (1992-2012) <Katrina/Rita Population Adjustment>-Linked To County Attributes-Total US, 1969-2013 Counties. Bethesda, MD: National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, Surveillance Systems Branch; 2015.
- Surveillance, Epidemiology, and End Results (SEER) Program. SEER*Stat Database: Incidence-SEER 18 Regs Research Data, Nov 2014 Sub (1992-2012) <Katrina/Rita Population Adjustment>-Linked To County Attributes-Total US, 1969-2013 Counties. Bethesda, MD: National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, Surveillance Systems Branch; 2015.
- DevCan: Probability of Developing or Dying of Cancer Software, Version 6.7.3. Bethesda, MD: Statistical Research and Applications Branch, National Cancer Institute; 2015.
- Howlander N, Noone AM, Krapcho M, et al, eds. SEER Cancer Statistics Review, 1975-2012 [based on November 2014 SEER data submission, posted to the SEER Web site April 2015]. Bethesda, MD: National Cancer Institute; 2015.
- Surveillance, Epidemiology, and End Results (SEER) Program. SEER*Stat Database: North American Association of Central Cancer Registries (NAACCR) Incidence-CiNA Analytic File, 1995-2012, Custom File With County, North American Association of Central Cancer Registries. Bethesda, MD: National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, Cancer Statistics Branch; 2015.
- Copeland G, Lake A, Firth R, et al. Cancer in North America: 2008-2012. Volume 1: Combined Cancer Incidence for the United States, Canada and North America. Springfield, IL: North American Association of Central Cancer Registries, Inc; 2015.
- Copeland G, Lake A, Firth R, et al. Cancer in North America: 2008-2012. Volume 2: Registry-Specific Cancer Incidence for the United States, Canada and North America. Springfield, IL: North American Association of Central Cancer Registries, Inc; 2015.
- Surveillance, Epidemiology, and End Results (SEER) Program. SEER*Stat Database: Mortality-All COD, Aggregated With State, Total US (1969-2012) <Katrina/Rita Population Adjustment>. Bethesda, MD: National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, Surveillance Systems Branch; 2015. Underlying mortality data provided by the National Center for Health Statistics.
- Surveillance Research Program, National Cancer Institute. SEER*Stat Software, Version 8.2.1. Bethesda, MD: National Cancer Institute; 2015.
- Fritz A, Percy C, Jack A, et al, eds. International Classification of Diseases for Oncology, 3rd ed. Geneva: World Health Organization; 2000.
- World Health Organization. International Statistical Classification of Diseases and Related Health Problems. 10th Revision. Volumes I-III. Geneva: World Health Organization; 2011.
- Joinpoint Regression Program, Version 4.2.0.2. June 2015; Statistical Research and Applications Branch, National Cancer Institute.
- Pickle LW, Hao Y, Jemal A, et al. A new method of estimating United States and state-level cancer incidence counts for the current calendar year. *CA Cancer J Clin*. 2007;57:30-42.
- Zhu L, Pickle LW, Ghosh K, et al. Predicting US- and state-level cancer counts for the current calendar year: part II: evaluation of spatiotemporal projection methods for incidence. *Cancer*. 2012;118:1100-1109.
- Chen HS, Portier K, Ghosh K, et al. Predicting US- and state-level cancer counts for the current calendar year: part I: evaluation of temporal projection methods for mortality. *Cancer*. 2012;118:1091-1099.
- National Center for Health Statistics. National Health Interview Survey, 2000, 2013 and 2014. Public-use data file and documentation. cdc.gov/nchs/nhis.htm. Accessed November 4, 2015.
- National Center for Health Statistics. National Health and Nutrition Examination Survey, 2013-14. Public-use data file and documentation. cdc.gov/nchs/nhanes/search/nhanes13_14.aspx. Accessed November 4, 2015.
- Eltom MA, Jemal A, Mbulaiteye SM, Devesa SS, Biggar RJ. Trends in Kaposi's sarcoma and non-Hodgkin's lymphoma incidence in the United States from 1973 through 1998. *J Natl Cancer Inst*. 2002;94:1204-1210.
- Centers for Disease Control and Prevention. HIV Surveillance Report, 2013 Vol 25. cdc.gov/hiv/library/reports/surveillance. Accessed July 17, 2015.
- Schistosomes, Liver Flukes and Helicobacter pylori. IARC Working Group on the Evaluation of Carcinogenic Risks to Humans. Lyon, 7-14, 1994. *IARC Monogr Eval Carcinog Risks Hum*. 1994;61:1-241.
- Karimi P, Islami F, Anandasabapathy S, Freedman ND, Kamangar F. Gastric cancer: descriptive epidemiology, risk factors, screening, and prevention. *Cancer Epidemiol Biomarkers Prev*. 2014;23:700-713.
- Greenberg AJ, Vachon CM, Rajkumar SV. Disparities in the prevalence, pathogenesis and progression of monoclonal gammopathy of undetermined significance and multiple myeloma between blacks and whites. *Leukemia*. 2012;26:609-614.
- Simard EP, Naishadham D, Saslow D, Jemal A. Age-specific trends in black-white disparities in cervical cancer incidence in the United States: 1975-2009. *Gynecol Oncol*. 2012;127:611-615.
- Jemal A, Center MM, Ward E. The convergence of lung cancer rates between blacks and whites under the age of 40, United States. *Cancer Epidemiol Biomarkers Prev*. 2009;18:3349-3352.
- Zeng C, Wen W, Morgans AK, Pao W, Shu XO, Zheng W. Disparities by race, age, and sex in the improvement of survival for major cancers: results from the National Cancer Institute Surveillance, Epidemiology, and End Results (SEER) program in the United States, 1990 to 2010. *JAMA Oncol*. 2015;1:88-96.
- Ward E, Halpern M, Schrag N, et al. Association of insurance with cancer care utilization and outcomes. *CA Cancer J Clin*. 2008;58:9-31.
- Bach PB, Schrag D, Brawley OW, Galaznik A, Yakren S, Begg CB. Survival of blacks and whites after a cancer diagnosis. *JAMA*. 2002;287:2106-2113.
- Shavers VL, Brown ML. Racial and ethnic disparities in the receipt of cancer treatment. *J Natl Cancer Inst*. 2002;94:334-357.

33. Smedley BD, Stith AY, Nelson AR, eds. *Unequal Treatment: Confronting Racial and Ethnic Disparities in Health Care*. Committee on Understanding and Eliminating Racial and Ethnic Disparities in Health Care. Washington, DC: National Academy Press; 2002.
34. Komenaka IK, Martinez ME, Pennington RE Jr, et al. Race and ethnicity and breast cancer outcomes in an underinsured population. *J Natl Cancer Inst*. 2010;102:1178-1187.
35. Kish JK, Yu M, Percy-Laurry A, Altekruse SF. Racial and ethnic disparities in cancer survival by neighborhood socioeconomic status in Surveillance, Epidemiology, and End Results (SEER) registries. *J Natl Cancer Inst Monogr*. 2014;2014:236-243.
36. Du XL, Lin CC, Johnson NJ, Altekruse S. Effects of individual-level socioeconomic factors on racial disparities in cancer treatment and survival: findings from the National Longitudinal Mortality Study, 1979-2003. *Cancer*. 2011;117:3242-3251.
37. Albain KS, Unger JM, Crowley JJ, Coltman CA Jr, Hershman DL. Racial disparities in cancer survival among randomized clinical trials patients of the Southwest Oncology Group. *J Natl Cancer Inst*. 2009;101:984-992.
38. Alexander DD, Waterbor J, Hughes T, Funkhouser E, Grizzle W, Manne U. African-American and Caucasian disparities in colorectal cancer mortality and survival by data source: an epidemiologic review. *Cancer Biomark*. 2007;3:301-313.
39. Newman LA, Griffith KA, Jatoi I, Simon MS, Crowe JP, Colditz GA. Meta-analysis of survival in African American and white American patients with breast cancer: ethnicity compared with socioeconomic status. *J Clin Oncol*. 2006;24:1342-1349.
40. Braithwaite D, Tammemagi CM, Moore DH, et al. Hypertension is an independent predictor of survival disparity between African-American and white breast cancer patients. *Int J Cancer*. 2009;124:1213-1219.
41. Tammemagi CM, Nerenz D, Neslund-Dudas C, Feldkamp C, Nathanson D. Comorbidity and survival disparities among black and white patients with breast cancer. *JAMA*. 2005;294:1765-1772.
42. Yancik R, Wesley MN, Ries LA, et al. Comorbidity and age as predictors of risk for early mortality of male and female colon carcinoma patients: a population-based study. *Cancer*. 1998;82:2123-2134.
43. Wallace TA, Martin DN, Ambis S. Interactions among genes, tumor biology and the environment in cancer health disparities: examining the evidence on a national and global scale. *Carcinogenesis*. 2011;32:1107-1121.
44. Murthy VH, Krumholz HM, Gross CP. Participation in cancer clinical trials: race, sex-, and age-based disparities. *JAMA*. 2004;291:2720-2726.
45. DeSantis CE, Fedewa SA, Goding Sauer A, Kramer JL, Smith RA, Jemal A. Breast cancer statistics, 2015: convergence of incidence rates between black and white women. *CA Cancer J Clin*. 2016;66:31-42.
46. Warner ET, Tamimi RM, Hughes ME, et al. Racial and ethnic differences in breast cancer survival: mediating effect of tumor characteristics and sociodemographic and treatment factors. *J Clin Oncol*. 2015;33:2254-2261.
47. Keegan TH, Kurian AW, Gali K, et al. Racial/ethnic and socioeconomic differences in short-term breast cancer survival among women in an integrated health system. *Am J Public Health*. 2015;105:938-946.
48. Wheeler SB, Reeder-Hayes KE, Carey LA. Disparities in breast cancer treatment and outcomes: biological, social, and health system determinants and opportunities for research. *Oncologist*. 2013;18:986-993.
49. Tammemagi CM. Racial/ethnic disparities in breast and gynecologic cancer treatment and outcomes. *Curr Opin Obstet Gynecol*. 2007;19:31-36.
50. van Ravestein NT, Schechter CB, Near AM, et al. Race-specific impact of natural history, mammography screening, and adjuvant treatment on breast cancer mortality rates in the United States. *Cancer Epidemiol Biomarkers Prev*. 2011;20:112-122.
51. Curtis E, Quale C, Haggstrom D, Smith-Bindman R. Racial and ethnic differences in breast cancer survival: how much is explained by screening, tumor severity, biology, treatment, comorbidities, and demographics? *Cancer*. 2008;112:171-180.
52. Menashe I, Anderson WF, Jatoi I, Rosenberg PS. Underlying causes of the black-white racial disparity in breast cancer mortality: a population-based analysis. *J Natl Cancer Inst*. 2009;101:993-1000.
53. Smith-Bindman R, Miglioretti DL, Lurie N, et al. Does utilization of screening mammography explain racial and ethnic differences in breast cancer? *Ann Intern Med*. 2006;144:541-553.
54. Press R, Carrasquillo O, Sciaccia RR, Giardina EG. Racial/ethnic disparities in time to follow-up after an abnormal mammogram. *J Womens Health (Larchmt)*. 2008;17:923-930.
55. Hoffman HJ, LaVerda NL, Levine PH, et al. Having health insurance does not eliminate race/ethnicity-associated delays in breast cancer diagnosis in the District of Columbia. *Cancer*. 2011;117:3824-3832.
56. Sail K, Franzini L, Lairson D, Du X. Differences in treatment and survival among African-American and Caucasian women with early stage operable breast cancer. *Ethn Health*. 2012;17:309-323.
57. Smith GL, Shih YC, Xu Y, et al. Racial disparities in the use of radiotherapy after breast-conserving surgery: a national Medicare study. *Cancer*. 2010;116:734-741.
58. Bickell NA, Wang JJ, Oluwole S, et al. Missed opportunities: racial disparities in adjuvant breast cancer treatment. *J Clin Oncol*. 2006;24:1357-1362.
59. Gorin SS, Heck JE, Cheng B, Smith SJ. Delays in breast cancer diagnosis and treatment by racial/ethnic group. *Arch Intern Med*. 2006;166:2244-2252.
60. Daly B, Olopade OI. A perfect storm: how tumor biology, genomics, and health care delivery patterns collide to create a racial survival disparity in breast cancer and proposed interventions for change. *CA Cancer J Clin*. 2015;65:221-238.
61. Keenan T, Moy B, Mroz EA, et al. Comparison of the genomic landscape between primary breast cancer in African American versus white women and the association of racial differences with tumor recurrence. *J Clin Oncol*. 2015;33:3621-3627.
62. Danforth DN Jr. Disparities in breast cancer outcomes between Caucasian and African American women: a model for describing the relationship of biological and nonbiological factors [serial online]. *Breast Cancer Res*. 2013;15:208.
63. Dietze EC, Sistrunk C, Miranda-Carboni G, O'Regan R, Seewaldt VL. Triple-negative breast cancer in African-American women: disparities versus biology. *Nat Rev Cancer*. 2015;15:248-254.
64. Blows FM, Driver KE, Schmidt MK, et al. Subtyping of breast cancer by immunohistochemistry to investigate a relationship between subtype and short and long term survival: a collaborative analysis of data for 10,159 cases from 12 studies [serial online]. *PLoS Med*. 2010;7:e1000279.
65. Perou CM, Borresen-Dale AL. Systems biology and genomics of breast cancer serial online. *Cold Spring Harb Perspect Biol*. 2011;3:a003293.
66. Newman LA. Disparities in breast cancer and African ancestry: a global perspective. *Breast J*. 2015;21:133-139.
67. Haddad SA, Lunetta KL, Ruiz-Narvaez EA, et al. Hormone-related pathways and risk of breast cancer subtypes in African American women. *Breast Cancer Res Treat*. 2015;154:145-154.
68. Palmer JR, Viscidi E, Troester MA, et al. Parity, lactation, and breast cancer subtypes in African American women: results from the AMBER Consortium [serial online]. *J Natl Cancer Inst*. 2014;106:dju237.
69. Phipps AI, Chlebowski RT, Prentice R, et al. Reproductive history and oral contraceptive use in relation to risk of triple-negative breast cancer. *J Natl Cancer Inst*. 2011;103:470-477.
70. Millikan RC, Newman B, Tse CK, et al. Epidemiology of basal-like breast cancer. *Breast Cancer Res Treat*. 2008;109:123-139.
71. Ursin G, Bernstein L, Lord SJ, et al. Reproductive factors and subtypes of breast cancer defined by hormone receptor and histology. *Br J Cancer*. 2005;93:364-371.
72. Yang XR, Sherman ME, Rimm DL, et al. Differences in risk factors for breast cancer molecular subtypes in a population-based study. *Cancer Epidemiol Biomarkers Prev*. 2007;16:439-443.
73. Doubeni CA, Major JM, Laiyemo AO, et al. Contribution of behavioral risk factors and obesity to socioeconomic differences in colorectal cancer incidence. *J Natl Cancer Inst*. 2012;104:1353-1362.
74. Irby K, Anderson WF, Henson DE, Devesa SS. Emerging and widening colorectal carcinoma disparities between blacks and whites in the United States (1975-2002). *Cancer Epidemiol Biomarkers Prev*. 2006;15:792-797.
75. Lansdorp-Vogelaar I, Kuntz KM, Knudsen AB, van Ballegooijen M, Zauber AG, Jemal A. Contribution of screening and survival differences to racial disparities in colorectal cancer rates. *Cancer Epidemiol Biomarkers Prev*. 2012;21:728-736.
76. Robbins AS, Siegel RL, Jemal A. Racial disparities in stage-specific colorectal cancer mortality rates from 1985 to 2008. *J Clin Oncol*. 2012;30:401-405.

77. White A, Vernon SW, Franzini L, Du XL. Racial disparities in colorectal cancer survival: to what extent are racial disparities explained by differences in treatment, tumor characteristics, or hospital characteristics? *Cancer*. 2010;116:4622-4631.
78. Du XL, Fang S, Vernon SW, et al. Racial disparities and socioeconomic status in association with survival in a large population-based cohort of elderly patients with colon cancer. *Cancer*. 2007;110:660-669.
79. Gomez SL, O'Malley CD, Stroup A, Shema SJ, Satariano WA. Longitudinal, population-based study of racial/ethnic differences in colorectal cancer survival: impact of neighborhood socioeconomic status, treatment and comorbidity [serial online]. *BMC Cancer*. 2007;7:193.
80. Polite BN, Dignam JJ, Olopade OI. Colorectal cancer model of health disparities: understanding mortality differences in minority populations. *J Clin Oncol*. 2006;24:2179-2187.
81. Potosky AL, Harlan LC, Kaplan RS, Johnson KA, Lynch CF. Age, sex, and racial differences in the use of standard adjuvant therapy for colorectal cancer. *J Clin Oncol*. 2002;20:1192-1202.
82. Dimou A, Syrigos KN, Saif MW. Disparities in colorectal cancer in African-Americans vs whites: before and after diagnosis. *World J Gastroenterol*. 2009;15:3734-3743.
83. Jessup JM, Stewart A, Greene FL, Minsky BD. Adjuvant chemotherapy for stage III colon cancer: implications of race/ethnicity, age, and differentiation. *JAMA*. 2005;294:2703-2711.
84. Yoon HH, Shi Q, Alberts SR, et al. Racial differences in BRAF/KRAS mutation rates and survival in stage III colon cancer patients [serial online]. *J Natl Cancer Inst*. 2015;107:djv186.
85. Anderson C, Burns DM. Patterns of adolescent smoking initiation rates by ethnicity and sex. *Tob Control*. 2000;9(suppl 2):II4-II8.
86. Steele CB, Pisu M, Richardson LC. Urban/rural patterns in receipt of treatment for non-small cell lung cancer among black and white Medicare beneficiaries, 2000-2003. *J Natl Med Assoc*. 2011;103:711-718.
87. Hardy D, Liu CC, Xia R, et al. Racial disparities and treatment trends in a large cohort of elderly black and white patients with nonsmall cell lung cancer. *Cancer*. 2009;115:2199-2211.
88. Farjah F, Wood DE, Yanez ND 3rd, et al. Racial disparities among patients with lung cancer who were recommended operative therapy. *Arch Surg*. 2009;144:14-18.
89. Ganti AK, Subbiah SP, Kessinger A, Gonsalves WI, Silberstein PT, Loberiza FR Jr. Association between race and survival of patients with non-small-cell lung cancer in the United States Veterans Affairs population. *Clin Lung Cancer*. 2014;15:152-158.
90. Zheng L, Enewold L, Zahm SH, et al. Lung cancer survival among black and white patients in an equal access health system. *Cancer Epidemiol Biomarkers Prev*. 2012;21:1841-1847.
91. Zullig LL, Carpenter WR, Provenza DT, et al. The association of race with timeliness of care and survival among Veterans Affairs health care system patients with late-stage non-small cell lung cancer. *Cancer Manag Res*. 2013;5:157-163.
92. Jemal A, Fedewa SA, Ma J, et al. Prostate cancer incidence and PSA testing patterns in relation to USPSTF screening recommendations. *JAMA*. 2015;314:2054-2061.
93. Mordukhovich I, Reiter PL, Backes DM, et al. A review of African American-white differences in risk factors for cancer: prostate cancer. *Cancer Causes Control*. 2011;22:341-357.
94. Rebbeck TR, Devesa SS, Chang BL, et al. Global patterns of prostate cancer incidence, aggressiveness, and mortality in men of African descent [serial online]. *Prostate Cancer*. 2013;2013:560857.
95. Center MM, Jemal A, Lortet-Tieulent J, et al. International variation in prostate cancer incidence and mortality rates. *Eur Urol*. 2012;61:1079-1092.
96. Fernandez P, Salie M, du Toit D, van der Merwe A. Analysis of prostate cancer susceptibility variants in South African men: replicating associations on chromosomes 8q24 and 10q11 [serial online]. *Prostate Cancer*. 2015;2015:465184.
97. Hsing AW, Yeboah E, Biritwum R, et al. High prevalence of screen detected prostate cancer in West Africans: implications for racial disparity of prostate cancer. *J Urol*. 2014;192:730-735.
98. Brawley OW, Ankerst DP, Thompson IM. Screening for prostate cancer. *CA Cancer J Clin*. 2009;59:264-273.
99. Chu KC, Tarone RE, Freeman HP. Trends in prostate cancer mortality among black men and white men in the United States. *Cancer*. 2003;97:1507-1516.
100. Cooperberg MR, Grossfeld GD, Lubeck DP, Carroll PR. National practice patterns and time trends in androgen ablation for localized prostate cancer. *J Natl Cancer Inst*. 2003;95:981-989.
101. Hankey BF, Feuer EJ, Clegg LX, et al. Cancer surveillance series: interpreting trends in prostate cancer—part I: evidence of the effects of screening in recent prostate cancer incidence, mortality, and survival rates. *J Natl Cancer Inst*. 1999;91:1017-1024.
102. Andriole GL, Crawford ED, Grubb RL 3rd, et al. Mortality results from a randomized prostate-cancer screening trial. *N Engl J Med*. 2009;360:1310-1319.
103. Hugosson J, Carlsson S, Aus G, et al. Mortality results from the Göteborg randomised population-based prostate-cancer screening trial. *Lancet Oncol*. 2010;11:725-732.
104. Schroder FH, Hugosson J, Roobol MJ, et al. Screening and prostate-cancer mortality in a randomized European study. *N Engl J Med*. 2009;360:1320-1328.
105. Schmid M, Meyer CP, Reznor G, et al. Racial differences in the surgical care of Medicare beneficiaries with localized prostate cancer. *JAMA Oncol*. 2016;2:85-93.
106. Moses KA, Paciork AT, Penson DF, Carroll PR, Master VA. Impact of ethnicity on primary treatment choice and mortality in men with prostate cancer: data from CaPSURE. *J Clin Oncol*. 2010;28:1069-1074.
107. Chornokur G, Dalton K, Borysova ME, Kumar NB. Disparities at presentation, diagnosis, treatment, and survival in African American men, affected by prostate cancer. *Prostate*. 2011;71:985-997.
108. Jayadevappa R, Chhatre S, Johnson JC, Malkowicz SB. Variation in quality of care among older men with localized prostate cancer. *Cancer*. 2011;117:2520-2529.
109. Brawley OW. The meaning of race in prostate cancer treatment. *JAMA Oncol*. 2016;2:93-94.
110. Rositch AF, Nowak RG, Gravitt PE. Increased age and race-specific incidence of cervical cancer after correction for hysterectomy prevalence in the United States from 2000 to 2009. *Cancer*. 2014;120:2032-2038.
111. Benard VB, Howe W, Royalty J, Helsel W, Kammerer W, Richardson LC. Timeliness of cervical cancer diagnosis and initiation of treatment in the National Breast and Cervical Cancer Early Detection Program. *J Womens Health (Larchmt)*. 2012;21:776-782.
112. Horner MJ, Altekruse SF, Zou Z, Wideroff L, Katki HA, Stinchcomb DG. US geographic distribution of prevaccine era cervical cancer screening, incidence, stage, and mortality. *Cancer Epidemiol Biomarkers Prev*. 2011;20:591-599.
113. Simard EP, Fedewa S, Ma J, Siegel R, Jemal A. Widening socioeconomic disparities in cervical cancer mortality among women in 26 states, 1993-2007. *Cancer*. 2012;118:5110-5116.
114. Brookfield KF, Cheung MC, Lucci J, Fleming LE, Koniaris LG. Disparities in survival among women with invasive cervical cancer: a problem of access to care. *Cancer*. 2009;115:166-178.
115. Coker AL, Du XL, Fang S, Eggleston KS. Socioeconomic status and cervical cancer survival among older women: findings from the SEER-Medicare linked data cohorts. *Gynecol Oncol*. 2006;102:278-284.
116. Gillison ML, Chaturvedi AK, Lowy DR. HPV prophylactic vaccines and the potential prevention of noncervical cancers in both men and women. *Cancer*. 2008;113:3036-3046.
117. Petrosky E, Bocchini JA Jr, Hariri S, et al. Use of 9-valent human papillomavirus (HPV) vaccine: updated HPV vaccination recommendations of the advisory committee on immunization practices. *MMWR Morb Mortal Wkly Rep*. 2015;64:300-304.
118. Reagan-Steiner S, Yankey D, Jeyarajah J, et al. National, regional, state, and selected local area vaccination coverage among adolescents aged 13-17 years—United States, 2014. *MMWR Morb Mortal Wkly Rep*. 2015;64:784-792.
119. Smith RA, Manassaram-Baptiste D, Brooks D, et al. Cancer screening in the United States, 2015: a review of current American Cancer Society guidelines and current issues in cancer screening. *CA Cancer J Clin*. 2015;65:30-54.
120. Saslow D, Solomon D, Lawson HW, et al. American Cancer Society, American Society for Colposcopy and Cervical Pathology, and American Society for Clinical Pathology screening guidelines for the prevention and early detection of cervical cancer. *Am J Clin Pathol*. 2012;137:516-542.